

What You Can't See: Contingent Thinking Failures in Dynamic Markets *

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Abstract

People make correct inferences from *observed* events but make mistakes when reasoning about *hypothetical* events. I study the implications in competitive markets. I define *Dynamic Cursed Expectations* (DCE), where agents treat events occurring at different future periods as independent. I build on it to define Dynamic Cursed Expectations *Equilibrium* (DCEE) and prove its existence for financial economies in which traders have CARA preferences, access to riskless storage, and signals that inform only about the next period's dividends. I apply DCEE to an asset pricing model and show that in DCEE agents underestimate the variance of risky assets, which under risk aversion leads to overvaluation relative to the Rational Expectations Equilibrium benchmark.

1 Introduction

Prices convey valuable information to market participants. In financial markets, a stock's price reflects the market's expectations for the underlying company, offering useful insights to traders. In energy markets, oil prices relay information about supply chain disruptions and geopolitical events. Fully exploiting the information in prices often requires *contingent*

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thinking. For example, when submitting a limit order, financial traders should consider the value of the asset conditional on the strike price being met.

Recent experimental literature documents failures of contingent thinking, showing that people often struggle to understand the implications of hypothetical events, but find it easier to make inferences from the same events once they have occurred.¹ Benchmark solution concepts fail to account for this phenomenon. Under Rational Expectations Equilibrium, agents learn correctly from the known history, contemplate the joint distribution of all future unknowns, and make an optimal plan for all contingencies. When a contingency is realized, their behavior matches their initial plan. In many real-world situations, this requires high levels of contingent thinking.

This paper introduces two new concepts that can serve as an alternative benchmark and trace out the implications of failures of contingent thinking in competitive markets. These failures are shown by the experimental literature to be systematic, significant, and robust across environments. *Dynamic Cursed Expectations* (DCE) posits that agents learn correctly from the observed history but misunderstand the joint distribution of future events. They treat events occurring at different future periods as independent from one another, thereby failing to foresee updating their beliefs in light of future information. In other words, they fail to anticipate learning from future events. *Dynamic Cursed Expectations Equilibrium* is a corresponding equilibrium concept, achieved when all agents optimize according to DCE, and prices clear the markets. This model captures the spirit of failures of contingent thinking documented in laboratory settings, embodying them in a competitive markets framework.

Anticipation of future learning is critical for decision-making in various contexts where expectations matter. For instance, when households choose between a fixed-rate and a variable-rate loan, they benefit from understanding the path dependency of future inflation. Similarly, when producers decide how much to invest in the production of a new good, they need to estimate future revenues. To optimize these decisions, agents must consider how to condition their actions on events that have yet to occur. For example, if a favorable shock in January reduces production costs, this will likely be associated with lower market prices in February, thus reducing February revenues, since other producers would also have benefited from the same shock. Grasping these interdependencies is challenging. To do so, agents must understand the full distribution of potential paths of decision-relevant variables. Once an event occurs, learning becomes more straightforward, as agents can focus on the paths following that realization and disregard those no longer relevant.

The experimental literature documents contingent thinking mistakes even in simple settings with a small number of contingencies. [Ngangoué and Weizsäcker \(2021\)](#) show this in

¹For a recent review of this literature, see [Niederle and Vespa \(2023\)](#).

a setting that is particularly relevant for market behavior. Subjects trade in a single risky asset that has a common value, where the price of the asset conveys information about its value. They compare a sequential treatment, where subjects are required to submit a limit order, and a simultaneous treatment, where subjects observe the price and then choose to buy or sell. Despite strategic equivalence between the treatments, 37% of traders fail to learn from the price in the sequential treatment, compared to only 19% in the simultaneous treatment. The setting of [Ngangoué and Weizsäcker \(2021\)](#) is quite simple: there are only two states of the world, two signals, and two time periods. These results suggest that in more complex financial markets with large number of contingencies, the bias is likely even stronger. Failures of contingent thinking also emerge in other economic environments, as documented by the experimental literature. In a voting game, agents fail to condition on being pivotal, but make correct inference from pivotality when it is known ([Esponda and Vespa, 2014](#)). In common value auctions, the winner’s curse is reduced in a Dutch (descending-price) auction relative to the strategically equivalent first-price auction ([Levin et al., 1996, 2016](#)). Similarly, the winner’s curse is weakened in second-price auctions when less contingent thinking is required ([Moser, 2019](#)). Studying belief updating directly, [Aina et al. \(2023\)](#) find that contingent thinking leads to significant deviations from Bayesian updating.

These empirical regularities highlight the need for new theoretical models. The evidence suggests that learning from observed events is easier than foreseeing learning from hypothetical ones. To model this in competitive markets, suppose that agents make correct inferences from everything they observe at the point of decision, but don’t anticipate making additional inferences based on hypothetical events. This paper presents a model in which agents observe all their fundamentals and private signals, up to time t , and all the past prices, up to time $t - 1$. Given this information, they submit demand correspondences (or limit orders), which determine the prevailing price. In particular, they fail to take into account the information content of prices that have not yet realized; this includes future prices along with present prices. Present prices are hypothetical at the time that agents declare their demand curves. Such individual behavior is captured by the definition of DCE.

DCE agents fail to learn from present prices, generating trade in settings where rational agents would not trade. Imagine a financial trader who learns that a company may be developing a new technology, leading to a potential increase in the value of the company’s stocks. A DCE trader submits a limit order, ordering to buy the stock if the price is lower than some $\bar{p}$, thinking that if the stock price is low enough, the investment is worthwhile. The trader’s reasoning misses one key aspect: when the price is below $\bar{p}$, this indicates that other investors think that the new technology is unlikely to be developed—the price reflects information held by others about the asset. The DCE agent trades, even though they can

only lose from trading. In contrast, a trader with Rational Expectations (RE) understands that the price fully reflects the market information, and that there is no potential gain from trade.

DCE agents learn from past prices, and therefore are not completely naive. When the same trader observes the low price, they learn that the price likely contains bad news about the development of the new technology. If they were given a chance to rescind the limit order, they would do so and not buy the stock.

While failure to learn from *present* prices leads to overtrading, failure to anticipate learning from *future* prices result in persistent overpricing of risky assets. Overtrading arises from heterogeneity in valuations among agents with different signals and is symmetric, whereas overpricing is asymmetric. To see why, consider a risk-averse trader who buys a risky asset on Monday, planning to either resell it on Tuesday or keep it and receive a dividend on Wednesday. The trader understands correctly the distribution of the price on Tuesday and that of the dividend on Wednesday, but not their joint distribution. If information is released over time, Tuesday’s price will be positively correlated with Wednesday’s dividend: when the resell price is high, the dividend is likely to also be high. A DCE trader neglects this correlation and in doing so underestimates the risk in the asset, perceiving more opportunities for gain than there actually are.

The application in Section 4 extends this intuition to settings with multiple assets and a finite time horizon of any length, showing that risk aversion under DCE leads to excess risk-taking. In the corresponding equilibrium, Dynamic Cursed Expectations Equilibrium (DCEE), assets are *overvalued* in expectation relative to Rational Expectations Equilibrium (REE), where agents correctly anticipate all future correlations, such as those between Tuesday’s price and Wednesday’s dividend. This overvaluation is larger for assets with higher variance and increases as the time horizon over which uncertain events will unfold lengthens. Finally, as in the earlier example, the failure to learn from present prices results in *overtrading*: there is trade in DCEE, whereas no trade occurs in REE. The volume of trade increases with the number of distinct signals and decreases when there is higher uncertainty about the dividend value.

The definitions of DCE and DCEE are motivated by recent work in game theory on failures of contingent thinking. In [Eyster and Rabin’s \(2005\)](#) *Cursed Equilibrium* (CE), players in a normal form game fail to account for the correlation between other players’ information and actions. Building on CE and inspired by the experimental findings on failures of contingent thinking, [Cohen and Li \(2022\)](#) introduce *Sequential Cursed Equilibrium* (SCE), applying the idea of cursedness to extensive games. A key feature of SCE is that players are cursed regarding unknown actions of opponents but learn correctly from observed actions. SCE

players are only “cursed” when contingent thinking is required.

DCE serves as a market parallel to [Cohen and Li \(2022\)](#). SCE players fail to recognize that other players learn from each other’s actions when these actions are unknown, treating them as independent from the information they hold.² Transitioning from strategic interactions to competitive markets brings important differences: while players in a game respond to actions of other players and the interactions between them, agents in competitive markets respond to prices and their interactions with other exogenous quantities. In a first step exploring the implications of cursedness for competitive markets, [Eyster et al. \(2019\)](#) introduce *Cursed Expectations Equilibrium* (CEE), a static equilibrium concept where agents completely neglect the informational content in prices. More recently, [Bayona and Manzano \(2024\)](#) study the implications of CEE in the presence of market power. DCEE is a dynamic analogue of CEE, defined for more general settings that allows agents to learn from past prices while still neglecting the informational content of present and future prices.

Section 2 defines Dynamic Cursed Expectations in a general setting that can be applied to study a variety of economic problems. In the primary setup, agents make a choice at each period $1, \dots, T$, where the period’s choice is constrained by the previous choice, present fundamentals and present prices. The market comprises a finite number of price-taking agents whose utilities depend only on their sequence of choices, which are constrained by prices and fundamentals. They do not observe others’ choices but only market prices. To justify the price-taking assumption, each agent can be thought of as representing a continuum of identical agents. Future fundamentals and prices are uncertain, and agents can receive exogenous information about them. They make plans of action for each future chain of events (future history), and have prior beliefs about the likelihood of those different histories. This setup builds on standard settings, such as that in [Stokey et al. \(1989\)](#).

RE and DCE agents differ in the probability distribution they assign to future events. At any point in time, agents hold beliefs about the future distribution of prices, fundamentals, and private information. Priors are assumed to match the exogenous ex-ante distribution of fundamentals and signals. After observing a certain history, both RE and DCE agents form the same posterior beliefs through Bayesian updating. However, they use these beliefs differently: RE agents evaluate the expected payoffs using the correct joint distribution of future prices and fundamentals, while DCE agents focus their attention on the marginal distribution at each future period and multiply them, failing to take into account the interdependencies between them. The DCE agents treat future events that occur in different future periods as independent.

²A game consists of *players* taking *actions*, while a market consists of *agents*, or sometimes *traders*, making *choices*.

One motivation for this modeling approach is the idea that agents rely heavily on empirical data to interpret the world. Rather than forming a theoretical model of the economy or anticipating equilibrium adjustments, they use empirical information about the distributions of prices and fundamentals, gathered either through personal experience or available data. For example, a first-time homebuyer might look at past housing prices and interest rates to inform their choices. While agents have access to accurate data, and therefore know the correct marginal distributions, they fail to account for temporal correlations within the data. The assumption that agents use the correct marginal distribution aligns with the data-driven approach and ties the modeler’s hands. The idea that agents don’t form theoretical models like economists do is consistent with recent experimental evidence. [Andre et al. \(2024\)](#) show that financial investors, including many financial professionals, use “mental models” that ignore equilibrium effects and endogenous prices adjustments.

Another motivation is the principle of computational simplicity: computing DCE is simpler and generally requires less cognitive effort than computing RE. As a lower-dimensional object, the product of the marginals is easier to grasp than the full joint distribution. While separating complex joint distributions into their marginals is psychologically plausible, DCE is one of multiple ways to do so. Agents could, for instance, neglect within-period future correlations, not just between-period ones. I focus on the latter for two reasons. First, I avoid arbitrarily choosing which within-period future correlations to neglect. Second, I focus specifically on failures to anticipate contingent *learning*: failures to condition on future variables when estimating later future variables.

In many settings, RE and DCE predict the same behavior. This is the case when optimal choices in dynamic markets do not require contingent thinking. Differences arise only when anticipated contingent learning is necessary for optimal decision-making, such as when fundamentals and prices are path-dependent, or when asymmetric information provides an opportunity to learn from prices. In the asset pricing application of section 4, anticipated contingent learning is important due to the correlation between resell prices and later fundamentals.

Section 3 defines equilibrium for a model of asset holdings in a financial economy. Equilibrium—whether REE or DCEE—is reached when markets clear and all agents optimize based on their expectations. REE consists of a pricing function that assigns prices to each history together with one plan for each agent, made at the beginning of time. Each agent’s plan maximizes RE given the pricing function, and prices clear the market at each history. In contrast, DCEE consists of a pricing function and a *sequence of plans* for each agent in every history. Agents optimize DCE at each history given the pricing function, and the pricing function clears the market at each history. RE agents follow through with their original

plan in each history, whereas DCE agents may adjust their plans in each history, potentially exhibiting dynamic inconsistency. REE prices clear the markets based on the original plans, whereas DCEE prices clear the markets only for the first step of any revised plan. As shown in Section 4, differences between REE and DCEE arise even when DCEE remains dynamically consistent, due to agents' misperception of the likelihood of future histories.

I prove that DCEE exists in financial economies when traders have CARA utility, access to riskless storage, and imminent private information—signals inform only about the next period's dividends. These assumptions play complementary roles. Imminent private information makes learning from past prices futile: traders do just as well learning from public dividends and their own signals. CARA preferences, together with the riskless asset, make asset demand independent of wealth, so that equilibrium prices depend only on current signals. With wealth effects, the cross-sectional wealth distribution becomes a hidden state variable that past prices partially reveal, so past prices forecast future prices even under imminent private information, and existence of DCEE for general risk-averse preferences remains an open question. The proof is constructive, building the equilibrium by backward induction on trading dates.

Even with imminent private information, REE may not exist. The potential nonexistence of REE, first discovered by [Green \(1977\)](#), arises from the dual role of prices: REE prices both clear markets and convey information, creating a feedback loop that may not converge. In DCEE, each price has a single role at any given time: past prices provide information, while present prices clear markets. This preserves the classic idea of learning from prices ([Hayek, 2013](#); [Muth, 1961](#)), but only in a partial sense. Subsection 3.3 offers an example where DCEE exists but there is no REE.

Learning from past but not from present prices offers a potential resolution to the Grossman-Stiglitz paradox ([Grossman and Stiglitz, 1980](#)). The Grossman-Stiglitz paradox argues that if agents acquire information at a cost and market prices fully reflect this information, then in a Rational Expectations Equilibrium (REE), agents would have no incentive to gather costly information—since they could simply infer it from the prices. However, if no agents are incentivized to acquire information, prices would no longer reflect any meaningful information. [Hellwig \(1982\)](#) addresses this paradox by showing that when agents learn from past prices but not from present ones, there is an incentive to acquire information at a cost. In Hellwig's model, agents are myopic, focusing only on the next period's returns while disregarding later periods. Further, utility is mean-variance and signals are normal.

The asset-pricing application connects to the classic speculative-resale logic of [Harrison and Kreps \(1978\)](#), but the mechanism and predictions are different. In Harrison and Kreps, heterogeneous expectations can generate a speculative premium because an investor values

the option to resell the asset later to someone with a more optimistic belief. The asymmetry between overvaluation and undervaluation is driven in part by the restriction that investors cannot short sell the asset. In the application below, the model is symmetric and short-selling is permitted. Overvaluation does not come from excluding pessimists from the market, but from a symmetric misperception of intertemporal risk. DCE traders correctly perceive the marginal distributions of future dividends and resale prices, but fail to account for their covariance. High resale prices occur precisely in states where later dividends are also high. Hence, this neglect leads traders to understate the variance of holding risky assets and therefore to overvalue them.

This distinction also yields a different prediction for trading volume. In the DCEE application, as in the static model of [Eyster et al. \(2019\)](#), per-trader volume rises with the number of independently informed trader types. The reason is that as more independent signals enter the price, the gap between a trader’s own signal and the aggregate signal reflected in the price becomes larger, and DCE traders continue to trade on that gap because they do not condition their demand schedules on the information contained in the present price. This comparative static contrasts with the agreeing-to-disagree branch of the Harrison–Kreps tradition. In those models, traders recognize that prices reveal others’ information, so as the number of independently informed traders grows, the information embedded in the price can swamp any individual signal. In the benchmark overconfidence and dismissiveness specifications studied by [Eyster et al. \(2019\)](#), per-trader volume converges to zero, and under standard parameter restrictions is decreasing in the number of traders.

The literature studies different types of failures to learn from prices and other information. [Bastianello and Fontanier \(2023\)](#) develop a theory of Partial Equilibrium Thinking, where agents learn from prices correctly about fundamentals, but fail to realize that others do the same. Other models study agents that learn from their information, but use misspecified models to do so (see [Esponda and Pouzo \(2016\)](#); [Fudenberg et al. \(2017\)](#); [Eliaz and Spiegel \(2020\)](#); [Heidhues et al. \(2021\)](#); [Molavi \(2022\)](#) and others). My behavioral assumptions differ conceptually from those in the aforementioned papers in that when forming expectations, agents they don’t use a (wrong) model of the economy or the behavior of others, but rather the empirical distribution of prices and fundamentals. In doing so, they neglect the correlation between different time period. A different strand of literature documents correlation neglect, but is not focused specifically on the differentiation between hypothetical and observed events ([Enke and Zimmermann, 2019](#); [Eyster and Weizsacker, 2010](#)).

The paper proceeds as follows. Section 2 introduces a general theoretical framework, and defines DCE in it. Section 3 focuses on a financial economy, defines DCEE in it, presents an example in which DCEE exists but REE does not, and proves that DCEE exists under

CARA preferences and imminent private information. Section 4 presents an asset pricing application and shows that in that application, DCEE leads to overvaluation of risky assets. Lastly, section 5 concludes.

2 Theoretical Framework

This section formally presents the two problems of interest, defining Rational Expectations (RE) and Dynamic Cursed Expectations (DCE) in a general setting.

2.1 General Setup

In this section, I define Rational Expectations and Dynamic Cursed Expectations for individual decision-makers, assuming a general information structure, arbitrary utility functions, and a finite time horizon. Throughout, decision-makers are price-takers, with prices treated as exogenous. Price formation and general equilibrium in a financial economy will be addressed in Section 3.

Let $(X, \mathcal{X})$ and $(Z, \mathcal{Z})$, be measurable spaces, where X represents the set of possible values for the *choice variable*, and Z represents the set of possible values for the *exogenous shock*. The price space, $P \subseteq X^*$, is a subspace of the dual space³, which assigns a cost to each choice, $x \in X$, in $\mathbb{R}$. Time is discrete and finite, $t = 1, 2, \dots, T$.

The market comprises a finite set of agents, indexed by $i \in I$, each associated with a measurable space $(\Omega_i, \mathcal{O}_i)$, from which agent i 's *private signals* are drawn. Agent i has utility function U_i and receives a sequence of private signals $\{\omega_{it}\}_{t=1}^T$, where $\omega_{it} \in \Omega_i$ is the signal at time t . A finite number of agents is assumed to avoid subtleties arising from a continuum of IID random variables (Judd, 1985). To abstract from strategic motives, each agent can be considered to represent a continuum of identical agents, all of equal measure.

Define the product spaces $(Z^t, \mathcal{Z}^t)$, $(P^t, \mathcal{P}^t)$, and $(\Omega_i^t, \mathcal{O}_i^t)$, for each time period $t = 1, 2, \dots, T$. I use z , p , and ω to refer to elements of Z^T , P^T , and Ω^T , respectively, where $z_t \in Z$, $p_t \in P$, and $\omega_{it} \in \Omega_i^t$ are the time t components of z , p , and ω_i . Let $(N, \mathcal{N}) = (Z \times P \times \Omega_i, \mathcal{Z} \times \mathcal{P} \times \mathcal{O}_i)$ be the product space, with $\eta_{it} \in N$ representing all the information available to agent i at time t : $\eta_{it} = (z_t, p_t, \omega_{it})$. Additionally, let $z^t = (z_1, \dots, z_t)$, $p^t = (p_1, \dots, p_t)$, $\omega_i^t = (\omega_{i1}, \dots, \omega_{it})$, and $\eta_i^t = (\eta_{i1}, \dots, \eta_{it})$ denote partial sequences.

Fundamentals and signals are exogenous, and together they fully characterize the *general state* of the economy. Prices are endogenous from the perspective of the economy but treated

³The dual space of X is the set of linear functionals mapping X to $\mathbb{R}$. For example, if $X = \mathbb{R}^m$, then the dual space is $P \subseteq \mathbb{R}^m$.

as exogenous from the individual decision-maker's viewpoint. Let $\hat{\Omega} = \times_{i \in I} \Omega_i$ be the space of agents' signals at any given time. The prior distribution of fundamentals and signals is given by the PDF $f : Z^T \times \hat{\Omega}^T \rightarrow \mathbb{R}_{\geq 0}$. Since signals can inform agents about future fundamentals, they affect agents' choices and prices. Asymmetric information creates an incentive to learn from prices, aligning with Hayek (2013).

A period $t < T$ *public partial history* is defined by $h_t = (z^t, p^t)$, while a *public terminal history* is the pair $h = (z, p) \in Z^T \times P^T$. Define $H_t = Z^t \times P^t$ and $H = Z^T \times P^T$. An agent i 's *private partial history* consists of the public partial history and the agent's past private signals, denoted $\eta_i^t = (z^t, p^t, \omega_i^t)$. Similarly, a *private terminal history* is $\eta_i = (z, p, \omega_i)$. Let $N_i^t = Z^t \times P^t \times \Omega_i^t$, and $N_i^T = Z^T \times P^T \times \Omega_i^T$. Private histories will simply be referred to as *histories* when the agent is clear from the context.

For clarity, consider the following example: two traders—one uninformed (trader A) and one informed (trader B)—trade in a single risky financial asset. There are two time periods, 0, and 1. In period 1, the asset pays a dividend $d \in D = \{10, 20\}$, drawn with equal probability. In period 0, the traders decide on their demand for the asset, $x_A(p_0), x_B(p_0) \in X = \{-1, 0, 1\}$, conditional on the present price. The informed trader, trader B , receives a signal fully revealing the dividend, $\omega_B = \{d\}$, while the uninformed trader, trader A , receives no information, $\omega_A = \emptyset$. Trader A 's only information is that of the prior distribution.

In this example, a public terminal history is of the form $h = ((0, d), (p, 0))$. At period 0, no dividend is paid, and at period 1, there is no more trade and the price is set to zero. A private terminal history for Trader B has the form $\eta_B = ((0, d), (p, 0), (d, \emptyset))$, and for trader A , $\eta_A = ((0, d), (p, 0), (\emptyset, \emptyset))$.

Building towards the definition of Dynamic Cursed Expectations, I define a refinement of history, a *morning history*. Each time period can be thought of as partitioned to different stages: at the beginning of each period, the “morning”, the fundamentals z_t and private information ω_{it} are revealed. Later in the period, agents make choices and then the price vector p_t is determined endogenously. The determination of prices in equilibrium is discussed in Section 3. Let $\hat{\eta}_i^t = \eta_i^t \setminus \{p_t\}$ be the *morning history*, representing the information that agent i learns from at the time of decision making. The morning history includes past prices, but not the present price.

Each of the sequences z , p , and ω_i plays a different role. The sequence of fundamentals, $z \in Z^T$ is exogenous and influences the flow payoff, which depends on z_t at time t . The sequence of prices, $p \in P^T$, also affects the flow payoff but is determined in equilibrium. Lastly, the sequence of signals, ω_i , affects expectation formation but does not influence the flow payoff directly.

Let B_i denote the *space of beliefs* over histories η_i , where the elements of B_i are probability

measures over terminal histories in N_i^T . Let $\bar{b}_i$ be agent i 's prior belief, with the marginal over exogenous variables matching the true distribution: the marginal of $\bar{b}_i$ on $Z^T \times \Omega_i^T$ coincides with that of f . The prior belief about prices matches the equilibrium price distribution, which will be discussed in the following section. Any partial history η_i^t gives rise to a Bayesian belief over terminal histories, represented by the probability measure $\bar{b}_i[\eta_i^t] \in B$. For $\eta_i^t = (z^t, p^t, \omega_i^t)$, the belief $\bar{b}_i[\eta_i^t]$ represents the posterior over public terminal histories after observing η_i^t and given the prior $\bar{b}_i$. Histories contradicting η_i^t are assigned zero probability; for $\eta_i' = (z', p', \omega_i') \in N_i$, where $\eta_i' \not\leq_t \eta_i^t$, we have $\bar{b}_i[\eta_i^t](\eta_i') = 0$.

2.2 Plans and Feasible Plans

When making a choice at any time period before T , agents also formulate contingent plans for their future choices based on different future histories, subject to a continuous recursive constraint $\Gamma_i : X \times Z \times P \rightarrow X$. I will first define plans made at period 0, and proceed to define plans made at different morning histories. Plans made at different morning histories are useful to describe a system in which plans may not be realized, where initial plans alone do not capture equilibrium behavior.

Definition 2.1. A **period 0 plan** is a value $\pi_{i0}^0 \in X$ and a sequence of measurable functions from private partial histories to choices $\pi_{ir}^0 : N_i^r \rightarrow X$, $r = 1, 2, \dots$.

The *state space* is defined to be X , and at period 0, there are initial, commonly known values $z_0 \in Z; p_0 \in P$.

Definition 2.2. A period 0 plan π_i^0 is **feasible from** $x_{i0} \in X$ if:

1. $\pi_{i0}^0 \in \Gamma_i[x_{i0}, z_0, p_0]$,
2. $\pi_{ir}^0((\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r)) \in \Gamma_i[\pi_{ir-1}^0(\tilde{z}^{r-1}, \tilde{p}^{r-1}, \tilde{\omega}_i^{r-1}), \tilde{z}_r, \tilde{p}_r]$ for all $(\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r) \in Z^r \times P^r \times \Omega_i^r$, $r = 1, 2, \dots T$.

These definitions resemble the concept of plans and feasible plans in Chapter 9 of [Stokey et al. \(1989\)](#). Such plans are formulated once at the beginning of time and are assumed to be followed thereafter. However, under Dynamic Cursed Expectations, agents may deviate from their initial plans. To capture this, I now define morning history-specific plans.

To illustrate, consider the example again: each trader starts with zero units of an asset, and their choice is restricted by $\Gamma_i = \{-1, 0, 1\}$, meaning they can either buy, sell, or hold the asset. A period 0 plan of agent $i = A, B$ consists of a single value $\pi_{i0}^0 = x_i^1 \in \mathbb{R}$. The plan is feasible if $\pi_{i0}^0 \in \{-1, 0, 1\}$, implicitly assuming that the agent has enough additional resources to buy a unit at each feasible price and cover potential losses from shorting.

Plans can also be made at later periods. Suppose $\hat{\eta}_i^t = (z^t, p^{t-1}, \omega_i^t) \in N_i^t$ is a morning history at time t , and let $r > t$. Define:

$$\mathcal{J}_{ir}(\hat{\eta}_i^t) = \{(\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r) \in N_i^r : \exists p_t \in P \text{ such that } (\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r) \mid_{\leq t} = (z^t, p^t, \omega_i^t)\}$$

to be the set of length r public partial histories that are compatible with $\hat{\eta}_i^t$.

Definition 2.3. A *morning history- $\hat{\eta}_i^t$ plan* consists of a measurable function $\pi_{it}[\hat{\eta}_i^t] : P \rightarrow X$ and a sequence of measurable functions $\pi_{ir}[\hat{\eta}_i^t] : \mathcal{J}_{ir}(\hat{\eta}_i^t) \rightarrow X$ for all $r = t+1, t+2, \dots$

Definition 2.4. Let $\hat{\eta}_i^t = (z^t, p^{t-1}, \omega_i^t)$ be a morning history of length t . A morning history- $\hat{\eta}_i^t$ plan $\pi_i^{\hat{\eta}_i^t}$ is **feasible from** $x_{it} \in X$, if:

1. $\pi_{it}[\hat{\eta}_i^t](p_t) \in \Gamma_i[x_{it}, z_t, p_t]$,
2. $\pi_{ir}[\hat{\eta}_i^t]((\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r)) \in \Gamma_i[\pi_{ir-1}[\hat{\eta}_i^t](\tilde{z}^{r-1}, \tilde{p}^{r-1}, \tilde{\omega}_i^{r-1}), \tilde{z}_r, \tilde{p}_r]$ for any $(\tilde{z}^r, \tilde{p}^r, \tilde{\omega}_i^r) \in \mathcal{J}_{ir}(\hat{\eta}_i^t)$ where $r = t+1, \dots, T$.

Feasible plans satisfy the constraint at each period but impose no additional restrictions on consistency or optimality regarding beliefs or utilities. Let $\hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ denote the set of morning history- $\hat{\eta}_i^t$ plans that are feasible from x_{it} . When the context is clear, I simply refer to these as *morning plans*.

The following assumption, adapted from [Stokey et al. \(1989\)](#)'s Assumption 9.1, ensures that $\hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ is non-empty.

Assumption 2.5. Γ_i is nonempty-valued and its graph is $(\mathcal{X} \times \mathcal{X} \times \mathcal{Z} \times \mathcal{P})$ -measurable. Additionally, Γ_i has a measurable selection, meaning there exists a measurable function $k : X \times Z \times P \rightarrow X$ such that $k(\tilde{x}, \tilde{z}, \tilde{p}) \in \Gamma_i(\tilde{x}, \tilde{z}, \tilde{p})$ for all $(\tilde{x}, \tilde{z}, \tilde{p}) \in X \times Z \times P$.

Under Assumption 2.5, $\hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ is non-empty for each morning history $\hat{\eta}_i^t$ and $x_{it} \in X$. The proof follows directly from Lemma 9.1 in [Stokey et al. \(1989\)](#).

An *economy* is characterized by the collection $(T, I, X, Z, P, f, (\Omega_i)_{i \in I}, (U_i)_{i \in I}, (\Gamma_i)_{i \in I})$.

2.3 Individual Optimization

Assume that agent i has a utility function $U_i : X^T \times Z^T \times P^T \rightarrow \mathbb{R}$, writing $U_i(x_i; z, p)$ for the payoff of the choice sequence x_i when fundamentals and prices follow (z, p) , and a prior belief $\bar{b}_i$. Further, assume that the agent observes the morning history $\hat{\eta}_i^t = (z^t, p^{t-1}, \omega_i^t)$ and remembers their history of choices $x_i^t = (x_{ir})_{r=1}^t$.

2.3.1 Rational Expectations

Let $p_t \in P$ be the present price. The corresponding history is $\eta_i^t = (z^t, p^t, \omega_i^t)$, comprised of a morning history $\hat{\eta}_i^t$ together with p_t . Under *Rational Expectations* (RE), the expected utility from a morning plan $\pi_i = \pi_i[\hat{\eta}_i^t] \in \hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ is computed based on the posterior belief, $\bar{b}_i[\eta_i^t]$:

$$EU_i^R(\pi_i, \eta_i^t) = \int_{\eta' | \leq t = \eta_i^t} U_i((x_i^t, \pi_i(\eta')) ; \eta') d\bar{b}_i[\eta_i^t](\eta') \quad (1)$$

Here, $\pi_i(\eta')$ denotes the sequence of choices dictated by π_i for periods $t + 1$ onward at the terminal history η' , for any $\eta' \in \mathcal{J}_{iT}(\hat{\eta}_i^t)$. When updating their beliefs, agents learn from the history η_i^t that includes the present price p_t .

2.3.2 Dynamic Cursed Expectations

Under *Dynamic Cursed Expectations* (DCE), agent i evaluates a plan by learning from past events but not from hypothetical ones. The agent updates their belief based on fundamentals, signals, and prices up to and including period $t - 1$ using Bayesian updating. In period t , they only learn from the exogenous variables z_t and ω_{it} and not from the present price p_t .

For any belief $b_i \in B_i$, let b_{il} denote the *marginal* of b_i on dimension l , b_{il} is a probability distribution over $Z \times P \times \Omega_i$. These marginals are used to capture the agent's neglect of correlations between events at different future periods.

Agent i uses the updated belief $\bar{b}_i[\hat{\eta}_i^t]$, learning from the morning history. In forming expectation, the agent relies on marginals $\bar{b}_{il}[\hat{\eta}_i^t]$ rather than the full joint distribution, thus fails to anticipate learning from future events. Suppose that the present price is p_t , and as in the RE case, let η_i^t be the corresponding history, comprised of morning history $\hat{\eta}_i^t$ together with p_t .

It is useful to interpret DCE as an *as-if* representation of behavior, rather than as a claim that agents consciously formulate a coherent probabilistic model of the economy. A DCE agent chooses the plan that would be optimal if future histories were weighted according to a distorted weighting rule. This rule preserves Bayesian learning from the observed morning history, but changes how the agent evaluates hypothetical continuations of that history. In particular, the agent uses the correct marginal distribution of each future period's events, conditional on what she has already observed, but evaluates future histories as if events occurring in different future periods were independent.

Thus, DCE is a restriction on choice behavior. The agent behaves as if she assigns distorted weights to possible future histories, and maximizes expected utility under those weights. The distortion does not concern the payoff associated with a given terminal history:

conditional on a terminal history, the agent correctly evaluates the payoff generated by the plan. The mistake concerns only the relative weights attached to different terminal histories. This captures the idea that agents understand what would happen if a contingency were realized, but fail to anticipate how observing one future contingency would affect their beliefs about later contingencies.

Suppose that the present price is p_t , and as in the RE case, let η_i^t be the corresponding history, comprised of morning history $\hat{\eta}_i^t$ together with p_t . The DCE utility from a plan $\pi_i = \pi_i[\hat{\eta}_i^t]$ at a present price of p_t is computed by applying this distorted weighting rule to terminal histories:

$$EU_i^{DC}(\pi_i, \eta_i^t) = \int_{\eta' | \leq_t = \eta_i^t} U_i((x_i^t, \pi_i(\eta')) ; \eta') d\left(\bigotimes_{r=t+1}^T \bar{b}_{ir}[\hat{\eta}_i^t]\right)(\eta'_{t+1}, \dots, \eta'_T) \quad (2)$$

Thus, under DCE, the present price enters the problem only mechanically. A trader submits a demand schedule, so the plan must specify what she would choose at each possible present price, and the realized price determines the relevant budget constraint. But when evaluating the plan, the trader does not treat the event that a particular price is realized as informative.

Let $o_i^R(x_i^t, \bar{b}_i[\eta_i^t]) \subseteq \hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ denote the set of plans that maximize Equation 1; these are the optimal plans under RE. Similarly, let $o_i^{DC}(x_i^t, \bar{b}_i[\hat{\eta}_i^t]) \subseteq \hat{\Pi}_i[\hat{\eta}_i^t](x_{it})$ denote the set of morning plans that maximize Equation 2; these are the optimal morning plans under DCE. A morning plan maximizes Equation 1 or 2 if it does so at every present price p_t at which a maximizer exists; with unbounded choice sets there may be prices at which the supremum is not attained, and equilibrium only ever evaluates plans at prices where it is (Section 3).

In the running example, assume that traders are risk neutral and have utility $U_i = x_i \cdot (d - p)$. Equilibrium is defined here for this example, while the general definition appears in Section 3. A Rational Expectations Equilibrium (REE) consists of a pair of prices, one for each state $d \in \{10, 20\}$, and a set of plans such that both traders maximize utility according to RE and the prices clear the market. A Dynamic Cursed Expectations Equilibrium (DCEE) is identical, except traders maximize utility according to DCE. Market clearing requires that at the prevailing price, either one trader sells while the other buys, or both traders do nothing.

There is no trade in REE. When $d = 20$, the informed trader wants to buy the asset for any price $p < 20$. When $d = 10$, the informed trader wants to sell the asset for any price $p > 10$. The uninformed trader can deduce the dividend from the market equilibrium, leading to no trade (assuming no trade occurs in cases of indifference).

In contrast, there is always trade in DCEE. The informed trader's demand remains as it

was under RE. The uninformed trader only considers the prior information and wants to buy if $p < 15$ and sell if $p > 15$, remaining indifferent if $p = 15$. When $d = 20$, the uninformed trader sells one unit to the informed at a price $15 < p < 20$, below the asset value. When $d = 10$, the uninformed trader buys one unit from the informed trader at a price $10 < p < 15$, above the asset value. The uninformed trader always loses.

2.3.3 Partially Dynamic Cursed Expectations

Under *Partially Dynamic Cursed Expectations* an agent fully learns from past events but only partially anticipates learning from hypothetical events. Under χ -partially dynamic cursed equilibrium, the agent uses an expectation that combines rational expectations and dynamic cursed expectations:

$$\begin{aligned} EU_i^{PDC}(\pi_i, \eta_i^t) &= \chi EU_i^{DC}(\pi_i, \eta_i^t) + (1 - \chi) EU_i^R(\pi_i, \eta_i^t) \\ &= \int_{\eta' | \leq t = \eta_i^t} U_i(x_i^t, \pi_i(\eta')) \left\{ \chi d \left(\bigotimes_{r=t+1}^T \bar{b}_{ir}[\hat{\eta}_i^t] \right) (\eta'_{t+1}, \dots, \eta'_T) + (1 - \chi) d\bar{b}_i[\eta_i^t](\eta') \right\} \end{aligned}$$

In computing the expected utility, the agent correctly evaluated the utility that would arise from the plan π_i under terminal history η' , but misperceives the probability of η' . She assigns a weight of $1 - \chi$ to the correct posterior, conditioning on history η_i^t , and a weight of χ to the belief that considers the information in the morning history, $\hat{\eta}_i^t$ but not the information in the present price p_t , and ignores the correlation between events in different future periods, while correctly understanding the marginals of these events.

2.4 Dynamic Consistency

DCE agents need not be dynamically consistent. At a morning history $\hat{\eta}_i^t$, a DCE agent evaluates future histories using the product of the future marginal distributions implied by the information currently observed. If a later morning history $\hat{\eta}_i^r$, $r > t$, is reached, the agent observes new fundamentals and private signals, and updates on them. The continuation problem at r may therefore differ from the continuation problem the agent anticipated at t . A plan that is optimal when made need not remain optimal after new information is observed.

For this reason, DCEE is defined using morning-history-specific plans. The equilibrium does not require agents to commit at period 0 to a single plan that remains optimal at every later history. Instead, at each morning history, agents choose a plan that is DCE-optimal given the information they currently observe, and market clearing is imposed on the current choices generated by these plans.

3 Equilibrium and Existence in a Financial Economy

In this section, I define a Dynamic Cursed Expectations Equilibrium (DCEE) and prove its existence. To achieve this, I narrow the focus and concentrate on a financial economy.

3.1 A Financial Economy with Storage

Trade in the K risky assets occurs at dates $t = 1, \dots, T - 1$; the dividend vector d_t and the signals $\hat{\omega}_t = (\omega_{it})_{i \in I}$ are realized at the beginning of each period $t = 1, \dots, T$, and the final price is $p_T = 0$. In addition to the risky assets, traders have access to a riskless asset at an exogenous net interest rate of zero, available in perfectly elastic supply.⁴ Trader i enters period t with risky holdings $x_{it} \in \mathbb{R}^K$ and storage $b_{it} \in \mathbb{R}$ (her holdings of the riskless asset), so that her wealth at the trading stage is

$$W_{it} = b_{it} + (p_t + d_t) \cdot x_{it},$$

and she chooses consumption $c_{it} \in \mathbb{R}$, new storage $b_{it+1} \in \mathbb{R}$, and new holdings $x_{it+1} \in \mathbb{R}^K$ subject to the budget identity

$$c_{it} + b_{it+1} + p_t \cdot x_{it+1} = W_{it}. \quad (3)$$

Positions and storage are unrestricted in sign (short sales and riskless borrowing are permitted), and consumption is unrestricted in sign as well.⁵

The initial conditions are $x_{i1} = e_i$ and $b_{i1} = 0$, and at $t = T$ the trader consumes her terminal wealth, $c_{iT} = W_{iT}$. The K risky-asset markets clear at every trading date: $\sum_i x_{it+1} = \sum_i e_i =: S$ for $t = 1, \dots, T - 1$. Traders can store and borrow freely; this is ensured by the elastic outside supply of the riskless asset. The constraint correspondence

$$\Gamma_i((b, x), d, p) = \{(c, b', x') \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^K : c + b' + p \cdot x' = b + (p + d) \cdot x\}$$

is nonempty and closed with measurable graph, so Assumption 2.5 holds. The price space is $P = \mathbb{R}^K$, and the dividend space is the compact box $D = [\underline{d}, \bar{d}]^K$ with $0 \leq \underline{d} \leq \bar{d} < \infty$ (Assumption 3.3 below restates this together with the distributional assumptions). Each trader

⁴This is a standard modeling device of the noisy rational expectations literature following Hellwig (1982).

⁵Unbounded consumption is the standard convention in CARA settings, where consumption is best read as a net monetary flow and the exponential penalty makes very negative realizations strongly disliked; see Hellwig (1982) and the treatment in ?. Equivalently, c_{it} may be read as consumption net of a baseline stream, with negative values meaning consumption below baseline. The application of Section 4 adopts the same convention: with normal dividends, monetary flows are negative with positive probability.

i has a continuous, strictly increasing, and concave utility function U_i over her consumption stream $(c_{i1}, \dots, c_{iT})$; the existence result of Subsection 3.4 further specializes to CARA utility. In the notation of Section 2, the period- t choice is (b_{it+1}, x_{it+1}) , with consumption determined residually by (3).

3.2 Defining REE and DCEE

Agents have asymmetric information and cannot engage in futures trading. Trade occurs sequentially, meaning that it cannot be fully contingent on all uncertainties. Therefore, equilibrium concepts such as those of Arrow and Debreu (1954) do not apply. If traders had the option to renegotiate, DCEE agents might benefit from new information, while Rational Expectations (RE) agents would be indifferent. Allowing some future contracts, while restricting trade to information known to both parties, would likely lead to similar results. However, this extension is left for future work.

In a Rational Expectations Equilibrium (REE), prices serve two roles: they both clear the market and convey information. From the perspective of a single agent, prices may reflect private information held by other agents, making learning from prices potentially valuable. In REE, agents fully understand the information embedded in present and future prices and the joint distribution of future prices, fundamentals, and private signals.

In contrast, in a Dynamic Cursed Expectations Equilibrium (DCEE), the two roles of prices are separated. Only past prices are used as information sources. Present and future prices are seen as constraints but not as learning tools. Agents fail to learn not only from future prices but also from any future information.

In the previous section, prices were treated as random variables. However, in equilibrium, whether REE or DCEE, period t prices are functions of observed prices, dividends, and private signals of all agents $\{\omega_{it}\}_i$.

Let $\rho = (\rho_t)_t$ denote a *pricing function*, where $\rho_t : D^t \times \hat{\Omega}^t \rightarrow P$ is measurable. This function maps partial sequences of dividends and private information to corresponding prices. Let Q be the set of pricing functions. Given ρ , trader i 's *price-belief* $b_{\rho,i}$ is the joint distribution of dividends, prices, and her own signals induced by the primitive distribution of $(d, \hat{\omega})$ together with $p_t = \rho_t(d^t, \hat{\omega}^t)$; her posterior at a history is the corresponding conditional distribution.

Definition 3.1. *A Rational Expectation Equilibrium consists of a pricing function ρ and a sequence of period-0 plans $(\pi_i^0)_i$ for each agent such that:*

1. *Agent i 's plan is rationally optimal given their beliefs $b_{\rho,i}$; that is, $\pi_i^0 \in o_i^R(\emptyset, b_{\rho,i})$.*

2. Agents' plans clear the market at each history. Specifically, given a time period t , a public history h^t , and the private signals ω_i^t for each i , the following holds: $\sum_i \pi_i^0(h^t, \omega_i^t) = \sum_i e_i$.

When the market is composed solely of DCE agents, we can turn to define *Dynamic Cursed Expectations Equilibrium* (DCEE). For each agent i , define a *sequence of morning plans* as a set of plans $(\pi_i^{\hat{\eta}_i^t})_{\hat{\eta}_i^t}$, where each plan corresponds to a morning history $\hat{\eta}_i^t$. The morning history is given by $\hat{\eta}_i^t = (d^t, p^{t-1}, \omega_i^t)$, and the plan is feasible based on the previous choice dictated by the morning history $\hat{\eta}_i^{t-1}$ plan: $\pi_i[\hat{\eta}_i^t] \in \hat{\Pi}_i[\hat{\eta}_i^t](\pi_{i,t-1}[\hat{\eta}_i^{t-1}](p_{t-1}))$. Let Π_i be the set of sequences of morning plans.

Definition 3.2. A *Dynamic Cursed Expectations Equilibrium* consists of a pricing function ρ and a sequence of morning-history-specific plans, $(\pi_i[\hat{\eta}_i^t])_{\hat{\eta}_i^t}$, for each agent i , such that:

1. At each history $\hat{\eta}_i^t$, and after a sequence of choices x_i^t , agent i 's plan is DCE-optimal based on her updated beliefs, which incorporates learning from the history $\hat{\eta}_i^t$. Formally, $\pi_i[\hat{\eta}_i^t] \in o_i^{DC}(x_i^t, b_{\rho,i}[\hat{\eta}_i^t])$.
2. At each ρ -consistent history $\hat{\eta}_i^t$ (defined below), with the present price $p_t = \rho_t(d^t, \hat{\omega}^t)$, present markets clear: $\sum_i \pi_{it}[\hat{\eta}_i^t](p_t) = \sum_i e_i$.

The structure of equilibrium differs between REE and DCEE. In an REE, a single initial plan is sufficient to describe the equilibrium, whereas in a DCEE, the equilibrium consists of a history-specific plan for each history. In an REE, when an agent at period 0 plans for choices at a future private partial history η_i^t , they correctly anticipate how their beliefs will evolve in that history. As a result, their period-0 plan for action at η_i^t is the same as their actual choice once η_i^t is observed.

In contrast, DCEE agents fail to anticipate how their beliefs will change over time. Though they learn from past events, their choices after observing a history $\hat{\eta}_i^t = \eta_i^t \setminus \{p_t\}$ might differ from their period-0 plans for that history.

Market clearing also differs between the two equilibria. In an REE, period-0 plans clear the market at every history. In a DCEE, plans are history-specific, and while markets clear at period t , market clearing is not guaranteed for future histories. Specifically, if $r > t$ and each agent i has a morning history $\hat{\eta}_i^r$, then the markets at period r clear based on the morning history- $\hat{\eta}_i^r$ plans. However, these plans may differ from those made at history $\hat{\eta}_i^t$ for actions at $\hat{\eta}_i^r$. Two conventions complete the definition. Market clearing is required at every ρ -consistent history, i.e. at histories whose recorded prices satisfy $p_r = \rho_r(d^r, \hat{\omega}^r)$ for all $r \leq t$; at histories with inconsistent price records, past markets did not clear and no price can clear the present one, so the clearing requirement is vacuous there. Similarly, a morning plan is required to be

optimal at every present price at which an optimal choice exists: with unrestricted positions there are prices at which none does, for example because the trader perceives an arbitrage and her expected utility has an unattained supremum. The equilibrium construction only ever evaluates demand at prices where the optimum exists.

3.3 An Example of REE Nonexistence

This subsection shows a simple example in which DCEE exists, but there is no REE. This is an adaptation of an example by [Kreps \(1977\)](#); [Allen et al. \(1998\)](#) to a financial economy.

The existence and nonexistence conditions of REE were extensively studied in the 1970s and 1980s, as reviewed by [Allen et al. \(1998\)](#). [Green \(1973\)](#) (later published in [Green \(1977\)](#)) introduced the first example for REE nonexistence. [Kreps \(1977\)](#) presented a simpler example, adapted below for a financial economy. [Green \(1977\)](#) and [Grossman \(1981\)](#) show that REE exists when trade can be fully contingent in the state of the world, even when this state is unknown to the traders. Existence problems arise when trade can only be contingent on the information available to both sides of the trade. [Radner \(1979\)](#) shows that with a finite state space, REE *generically* exists—nonexistence only occurs for a negligible set of parameter values. [DeMarzo and Skiadas \(1998\)](#) introduce conditions for the existence of REE that is partially informative in a static model.

In the example, there are three time periods and two financial traders, A and B , that have utility from consumption,

$$U_A(c_A) = \log(c_{A2}) + 2\log(c_{A3})$$

$$U_B(c_B) = 2\log(c_{B2}) + \log(c_{B3})$$

Agent A prefers consumption at period 3 over consumption at period 2, while agent B prefers consumption at period 2 over consumption at period 3.

They trade in two assets, α and β , in two equally probable states of the world, s_1 and s_2 . The dividend payoffs at the two states are shown in Table 1. The dividends are a mirror image of each other: in state s_1 , α pays only at period 2 and β pays only at period 3, and the reverse in state s_2 . There is no utility from consumption at period 1.

	$d_2(\alpha)$	$d_3(\alpha)$	$d_2(\beta)$	$d_3(\beta)$
s_1	1	0	0	1
s_2	0	1	1	0

Table 1: Dividend pay for assets α, β at states s_1, s_2

In this example, assets are traded only at period 1; markets are closed thereafter, and there is no riskless asset (none is needed, since trade occurs once). Each trader initially holds three units of each asset, $e_A = e_B = (3, 3)$. At period 1, trader B knows the state and trader A does not. The economy has imminent private information: trader B 's signal is a function of the period-2 dividends.

3.3.1 To find the DCEE, solve backwards.

At period 3 there is no trade, and consumption is based on existing ownership,

$$c_{A3} = x_{A3}(\alpha) \cdot d_3(\alpha) + x_{A3}(\beta) \cdot d_3(\beta)$$

$$c_{B3} = x_{B3}(\alpha) \cdot d_3(\alpha) + x_{B3}(\beta) \cdot d_3(\beta)$$

At period 2, the dividends d_2 are paid and all uncertainty is resolved; markets are closed, so $x_{A3} = x_{A2}$ and $x_{B3} = x_{B2}$.

It remains to find period 2's choices, x_{A2}, x_{B2} . period 2 consumption is:

$$c_{B2} = x_{B2}(\alpha) \cdot d_2(\alpha) + x_{B2}(\beta) \cdot d_2(\beta)$$

And is subject to the period 1 budget constraint:

$$p_\alpha x_{B2}(\alpha) + p_\beta x_{B2}(\beta) = 3p_\alpha + 3p_\beta \quad (4)$$

Suppose that the state is s_1 . Trader B solves:

$$\max_{x_{B2}, x_{B3}} 2 \log(x_{B2}(\alpha)) + \log(x_{B2}(\beta))$$

subject to 4. Demand is:

$$x_{B2}(\alpha) = 2 \frac{p_\alpha + p_\beta}{p_\alpha}$$

$$x_{B2}(\beta) = \frac{p_\alpha + p_\beta}{p_\beta}$$

When the state is s_2 , the demand for α and β is reversed. Trader A has no private information, and thus has symmetric demand of the two assets.

$$x_{A2}(\alpha) = \frac{3}{2} \frac{p_\alpha + p_\beta}{p_\alpha}$$

$$x_{A2}(\beta) = \frac{3}{2} \frac{p_\alpha + p_\beta}{p_\beta}$$

period 2 prices are set to clear the market. When the state is s_1 , the price ratio is: $p_\beta = \frac{5}{7}p_\alpha$, and when the state is s_2 , it is reversed, $p_\alpha = \frac{5}{7}p_\beta$.

3.3.2 What about REE?

In REE, DCEE prices cannot be an equilibrium: prices are different in each state, allowing trader A to learn the state from the prevailing price.

It turns out that a fully revealing equilibrium, where A known the state, is also impossible: in that case, A 's demand is a mirror image of B 's demand due to symmetry. Then, the prices would satisfy $p_\alpha = p_\beta$ in both states, making it impossible for A to learn the state!

3.4 Existence of DCEE

This subsection states the existence result. Two assumptions are imposed, one on information and one on preferences. The informational assumption is *imminent private information*, under which signals inform only about the next period's dividends. The preference assumption is that of constant absolute risk aversion. The discussion following the theorem explains what each assumption contributes and in what sense both are necessary for the construction.

Assumption 3.3 (Dividends). *Dividends take values in $D = [\underline{d}, \bar{d}]^K$ with $0 \leq \underline{d} \leq \bar{d} < \infty$; the vectors $d_1, \dots, d_T$ are i.i.d. across time, and ν denotes their common distribution on D , which is not supported in any affine hyperplane of $\mathbb{R}^K$. (ν need not have full support.)*

Assumption 3.4 (Imminent private information). *Signals inform only about next period's dividends: $\omega_{it} = g_i(d_{t+1}, \varepsilon_{it})$, where g_i is measurable, the noise terms ε_{it} take values in $\mathbb{R}^K$ and are i.i.d. across i and t , and the whole noise array is independent of the dividend process. (The noise is not required to be bounded.) Moreover, the signal has a conditional density $f_i(\omega | d)$ on $\mathbb{R}^K$ (its likelihood function), jointly continuous in (ω, d) , with $f_i(\omega | d) > 0$ for every signal value ω and every dividend value d in the support of ν .*

Assumption 3.3 is a regularity condition on fundamentals: dividends are bounded, and no linear combination of them is deterministic. Assumption 3.4 is the informational assumption: signals reach only one period ahead. Its positivity clause ensures that signals move likelihoods, never supports. This, together with Assumption 3.3, ensures that all traders' posteriors over d_{t+1} share the same support, one not contained in any affine hyperplane; its continuity clause makes posteriors, and with them equilibrium prices, vary continuously with the signals.

Assumption 3.5. *Trader i has CARA utility over consumption streams:*

$$U_i(c_{i1}, \dots, c_{iT}) = - \sum_{t=1}^T \delta_{it} e^{-\gamma_i c_{it}}, \quad \gamma_i > 0, \delta_{it} > 0.$$

As in the definition of DCEE, at a morning history $\hat{\eta}_i^t = (d^t, p^{t-1}, \omega_i^t)$ the trader updates on the morning history and evaluates plans by the product of the per-period marginals of her posterior over (d_r, p_r, ω_{ir}) , $r > t$.

Theorem 3.6. *Under Assumptions 3.3, 3.4, and 3.5, a Dynamic Cursed Expectations Equilibrium exists. Its pricing functions satisfy $\rho_t(d^t, \hat{\omega}^t) = \varrho_t(\hat{\omega}_t)$ for measurable maps ϱ_t of the current signals only, with values in $[(T-t)\underline{d}, (T-t)\bar{d}]^K$.*

Both assumptions are substantive, and they play complementary roles. Imminent private information makes past prices uninformative: everything a past price could reveal about the future is already revealed by observed dividends. CARA utility, together with the riskless asset, makes each trader's risky demand independent of her wealth, so that equilibrium prices depend only on current signals. With wealth effects, the cross-sectional wealth distribution would be a hidden state variable that past prices partially reveal, and past prices would then forecast future prices even under imminent private information. Existence of DCEE for general risk-averse preferences remains an open question. The proof is constructive: it builds the equilibrium by backward induction on trading dates, with each date's market cleared through an auxiliary planner problem whose shadow price is the equilibrium price. The asset-pricing application of Section 4 shares the structure assumed here—i.i.d. dividends, signals about next period's dividends, a safe asset in elastic supply—but is not covered by the theorem, for two reasons discussed there; its equilibrium is constructed explicitly.

4 Application: Asset Overvaluation with Mean-Variance Preferences

This section presents a financial model in which a Dynamic Cursed Expectations Equilibrium leads to overvaluation of risky assets relative to a Rational Expectations Equilibrium. Notably, the overvaluation is larger for assets with high variance. While there is no trade in the REE, trade occurs in the DCEE. DCEE trade is driven by the dispersion of private noise: the expected volume of trade increases with the number of traders and decreases with dividend uncertainty. Subsection ?? adds an atomistic rational trader to the cursed market and quantifies how much rationality is worth there.

4.1 Application Setup

Trade in financial assets occur periodically, whereas consumption only occurs at the very end. This setup is designed to reflect the behavior of financial traders who invest over long periods but frequently buy and sell financial assets, far more often than they withdraw their investments. The choice space is $X = \mathbb{R}^K$, the dividend space is $D = \mathbb{R}^K$, and the price space is $P = \mathbb{R}^K$.

The model involves agents who trade in potentially risky assets at periods $t = 0, \dots, T - 1$, with K such assets, each indexed by k . Dividends are distributed at each trading period $1, \dots, T$, and are drawn from a normal distribution: $d_t(k) \in \mathcal{N}(\mu_k, \sigma_k^2)$, IID across time. There is a safe asset in infinite supply, that allows for saving with no interest.

There are $N \geq 2$ agents, indexed by $i \in I = \{1, \dots, N\}$. Each agent i receives a signal vector s_t^i that provides information about d_{t+1} ; specifically, $s_t^i = d_{t+1} + \varepsilon_{it}$, where $\varepsilon_{it}(k) \in \mathcal{N}(0, \tau_k^2)$ are independent across agents and time. The vectors s_t^i , d_t , and ε_{it} have length K , with each coordinate represents the corresponding k 'th asset. Agents are assumed to be price-taking. To justify ignoring strategic motives, each agent i can be thought of as representing a continuum of identical agents, with a measure of 1 for each type, and N such types. This environment mirrors the setting of Theorem 3.6: dividends are i.i.d. across periods; signals inform only about next period's dividends, their additive Gaussian noise gives a positive, continuous likelihood, and the safe asset plays the role of the riskless storage. It differs in two respects: traders have mean-variance rather than CARA preferences, and dividends are normal, so their support is unbounded rather than a compact box. Theorem 3.6 therefore does not apply directly; existence for this application is instead established constructively below, by exhibiting the equilibrium in closed form.

Agents have mean-variance preferences, a common assumption in finance that is sometimes justified by a Taylor approximation of a general utility function. The utility of agent i at period t is given by:

$$U^i(c_i) = E_t^i[c_i] - \frac{\alpha}{2} \text{Var}_{it}(c_i)$$

After observing a morning history $\hat{\eta}_i^t = (z^t, p^{t-1}, \omega_i^t)$, trader i 's demand is $x_{it+1}(\hat{\eta}_i^t, p_t)$ at the price vector p_t .

In equilibrium, the price p_t is set to clear the market. Suppose that each agent is initially endowed with 1 unit of each asset. The total supply is $N \cdot \mathbf{1}$, where $\mathbf{1}$ is a vector of ones. The distribution of initial ownership is irrelevant for the results, as mean-variance preferences eliminate wealth effects. However, positive supply of the assets is crucial to obtain the main result of this section. A price p_t^* is part of an equilibrium if $\sum_i x_{it+1}(\hat{\eta}_i^t, p_t^*) = \mathbf{1}$. I will omit the dependency on $\hat{\eta}^t$ when the context is clear.

Traders' monetary flows are as follows:

- At period 0: $m_{i0} = -p_0 \cdot (x_{i1} - e_i)$
- At period $0 < t < T$: $m_{it} = d_t \cdot x_{it} - p_t \cdot (x_{it+1} - x_{it})$
- At period T : $m_{iT} = d_T \cdot x_{iT}$ (setting $p_T = 0$)

Rational Expectations (RE) agents and Dynamic Cursed Expectations (DCE) agents differ in how they interpret random variables, specifically in the probabilities they assign to future histories. Some of the analysis is identical for both, so I use E_t^i , Cov_t^i , Var_t^i to denote the corresponding operators. The specific solutions for REE and DCEE are presented in Subsections 4.2 and 4.3.

At period t , trader i solves:

$$E_t^i [u(m_{i0} + \dots + m_{iT})]$$

which can be written as:

$$E_t^i \left[u \left(p_0 \cdot e_i + (d_1 + p_1 - p_0) \cdot x_{i1} + (d_2 + p_2 - p_1) \cdot x_{i2} \right. \right. \\ \left. \left. + \dots + (d_{T-1} + p_{T-1} - p_{T-2}) \cdot x_{iT-1} + (d_T - p_{T-1}) \cdot x_{iT} \right) \right]$$

choosing x_{it+1} as well as a contingent plan for future holdings that depends on the realized future history.

At period t , when x_{it+1} is chosen, the expected utility becomes:

$$\begin{aligned} & E_t^i [c_i] - \frac{\alpha}{2} \text{Var}_t^i (c_i) \\ &= p_0 \cdot e_i + \sum_{r=1}^t (d_r + p_r - p_{r-1}) \cdot x_{ir} + (E[d_{t+1} + p_{t+1}] - p_t) \cdot x_{it+1} + \sum_{r=t+2}^T E_t^i [(d_r + p_r - p_{r-1}) \cdot x_{ir}] \\ &\quad - \frac{\alpha}{2} \text{Var}_t^i \left((d_{t+1} + p_{t+1} - p_t) \cdot x_{it+1} + \sum_{r=t+2}^T (d_r + p_r - p_{r-1}) \cdot x_{ir} \right) \quad (5) \end{aligned}$$

where the history up to period t is constant at this time.

$$\begin{aligned}
& \text{Var}_t^i \left((d_{t+1} + p_{t+1} - p_t) \cdot x_{it+1} + \sum_{r=t+2}^T (d_r + p_r - p_{r-1}) \cdot x_{ir} \right) \\
&= \sum_k (x_{it+1}(k))^2 \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k) - p_t(k)) + \sum_{r=t+2}^T \text{Var}_t^i ((d_r + p_r - p_{r-1}) \cdot x_{ir}) \\
&\quad + 2 \sum_{r \neq r' > t} \text{Cov}_t^i ((d_r + p_r - p_{r-1}) \cdot x_{ir}, (d_{r'} + p_{r'} - p_{r'-1}) \cdot x_{ir'})
\end{aligned}$$

Taking the first order condition of Equation 5 with respect to $x_{it+1}(k)$,

$$\begin{aligned}
& E_t^i [d_{t+1}(k) + p_{t+1}(k) - p_t(k)] \\
&\quad - \frac{\alpha}{2} \left(2x_{it+1}(k) \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k) - p_t(k)) \right. \\
&\quad \left. + 2 \sum_{r>t+1} \text{Cov}_t^i (d_{t+1}(k) + p_{t+1}(k) - p_t(k), (d_r(k) + p_r(k) - p_{r-1}(k)) x_{ir}(k)) \right) = 0 \quad (6)
\end{aligned}$$

At time t , x_r^i for $r > t + 1$ is viewed as a random variable that depends on the probabilities assigned to future histories. Agents are consistent in their planning, that is, their plan for choice at history η^{r-1} is the same as their actual choice when history η^{r-1} is realized, but the DCE agent may be wrong in the probability she assigns to each history. The present price p_t is realized at period t , and d_{t+1} is independent of d_r, p_{r-1} for $r > t + 1$. Equation 6 can be written as:

$$\begin{aligned}
& E_t^i [d_{t+1}(k) + p_{t+1}(k) - p_t(k)] \\
&\quad - \alpha \sum_{r>t+1} \text{Cov}_t^i (d_{t+1}(k) + p_{t+1}(k), (d_r(k) + p_r(k) - p_{r-1}(k)) x_{ir}(k)) \\
&\quad = \alpha x_{it+1}(k) \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k) - p_t(k))
\end{aligned}$$

Thus,

$$\begin{aligned}
x_{it+1}(k) &= \frac{E_t^i [d_{t+1}(k) + p_{t+1}(k)] - p_t(k)}{\alpha \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k))} \\
&\quad - \frac{\alpha \sum_{r>t+1} \text{Cov}_t^i (d_{t+1}(k) + p_{t+1}(k), (d_r(k) + p_r(k) - p_{r-1}(k)) x_{ir}(k))}{\alpha \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k))}
\end{aligned}$$

Note that since all traders' signals are drawn from the same distribution independently, $\text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k) - p_t(k))$ is independent of i and can be written as Var_t ; the realization

of the signal does not affect the posterior variance. Similarly, Cov_t^i can be written as Cov_t . Breaking the Covariance term into parts, most of its terms disappear. Thus, demand can be simplified to:

$$x_{it+1}(k) = \frac{E_t^i [d_{t+1}(k)] - p_t(k) + E_t^i [p_{t+1}(k)]}{\alpha \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k))} - \frac{\alpha \text{Cov}_t^i (d_{t+1}(k) + p_{t+1}(k), (d_{t+2}(k) + p_{t+2}(k) - p_{t+1}(k)) x_{it+2}(k))}{\alpha \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k))}$$

In equilibrium, market clearing implies that $\sum_i x_{it+1}^i(k) = N$ for all k :

$$\begin{aligned} \sum_i E_t^i [d_{t+1}(k) + p_{t+1}(k) - p_t(k)] \\ - \alpha \sum_i \sum_{r>t+1} \text{Cov}_t (d_{t+1}(k) + p_{t+1}(k), (d_r(k) + p_r(k) - p_{r-1}(k)) x_r^i(k)) \\ = \alpha N \text{Var}_t (d_{t+1}(k) + p_{t+1}(k) - p_t(k)) \end{aligned} \quad (7)$$

switching the order of summation, and using the additivity of covariance:

$$\begin{aligned} \sum_i \sum_{r>t+1} \text{Cov}_t (d_{t+1}(k) + p_{t+1}(k), (d_r(k) + p_r(k) - p_{r-1}(k)) x_{ir}(k)) \\ = \sum_{r>t+1} \text{Cov}_t \left(d_{t+1}(k) + p_{t+1}(k), (d_r(k) + p_r(k) - p_{r-1}(k)) \sum_i x_{ir}(k) \right) \\ = N \sum_{r>t+1} \text{Cov}_t (d_{t+1}(k) + p_{t+1}(k), d_r(k) + p_r(k) - p_{r-1}(k)) \end{aligned}$$

where the last equality uses market clearing again. The price can be extracted from 7 to be:

$$p_t(k) = \frac{1}{N} \sum_i E_t^i [d_{t+1}(k) + p_{t+1}(k)] - \alpha \sum_{r>t} \text{Cov}_t (d_{t+1}(k) + p_{t+1}(k), d_r(k) + p_r(k) - p_{r-1}(k)) \quad (8)$$

Prices at period t can thus depend *only* on:

- The realized dividend next period, d_{t+1} through private signals about it.
- The distribution of future dividends, d_r for $r > t + 1$, which is identical to the ex-ante distribution of those dividends. This is because the information available at period t is about dividends up to d_{t+1} .

- The distribution of future prices, p_r for $r > t$, which is identical to the ex-ante distribution of those prices.
- The joint distribution of those future dividends and prices.

Simplifying 8,

$$\begin{aligned}
p_t(k) &= \frac{1}{N} \sum_i E_t^i [d_{t+1}(k)] + E_t [p_{t+1}(k)] \\
&\quad - \alpha \sum_{r>t+1} \text{Cov}_t(p_{t+1}(k), d_r(k) + p_r(k) - p_{r-1}(k)) - \alpha \text{Var}_t(d_{t+1}(k)) - \alpha \text{Var}_t(p_{t+1}(k)) \\
&= \frac{1}{N} \sum_i E_t^i [d_{t+1}(k)] + E_t [p_{t+1}(k)] - \alpha \{ \text{Cov}_t(p_{t+1}(k), d_{t+2}(k)) + \text{Var}_t(d_{t+1}(k)) \}
\end{aligned}$$

I turn to define the volume of trade between agents. The volume of trade in asset k of agent i at period $t+1$ is defined to be the norm of change in asset holdings between x_{it+1} and x_{it+2} , $|x_{it+2}(k) - x_{it+1}(k)|$. The total trading volume is defined to be the sum of trading volumes for all agents,

$$V_{t+1}(k) = \sum_i |x_{it+2}(k) - x_{it+1}(k)|$$

and the expected per-agent trading volume is the expected trade generated by a single trader,

$$v_{t+1}(k) = E_{<t} [|x_{it+2}(k) - x_{it+1}(k)|]$$

4.2 Rational Expectations Equilibrium

Suppose that all traders use RE. Under REE, agents learn from their signals as well as present prices,

$$p_t(k) = \frac{1}{N} \sum_i E [d_{t+1}(k) | p_t, s_t^i] + E [p_{t+1}(k)] - \alpha \{ \text{Cov}(p_{t+1}(k), d_{t+2}(k)) + \text{Var}(d_{t+1}(k) | p_t) \} \quad (9)$$

for all t , where $E [p_{t+1}(k) | p_t, s_t^i] = E [p_{t+1}(k)]$ is the same as the ex-ante expectation; $\text{Cov}(p_{t+1}(k), d_{t+2}(k))$ is independent of both p_t and s_t^i , and the variance $\text{Var}(d_{t+1}(k) | p_t, s_t^i)$'s dependence on the private signal s_t^i is fully reflected in p_t .

To show that a REE exists in this application, guess a symmetric linear solution (e.g., [Eyster et al. \(2019\)](#)):

$$p_t(k) = \sum A_R(k) s_t^i(k) + B_R(k) \mu_k + E_t [p_{t+1}(k)] - \alpha \{ \text{Cov}_t(p_{t+1}(k), d_{t+2}(k)) + \text{Var}_t(d_{t+1}(k) | p_t) \}$$

And recall the general formula:

$$x_{it+1}(k) = \frac{E_t^i [d_{t+1}(k)] - p_t(k) + E_t^i [p_{t+1}(k)] - \alpha \text{Cov}_t^i(p_{t+1}(k), (d_{t+2}(k) + p_{t+2}(k) - p_{t+1}(k)) x_{it+2}(k))}{\alpha \text{Var}_t^i(d_{t+1}(k) + p_{t+1}(k))}$$

From an informational point of view of agent i , $p_t(k)$ can be thought of as a signal,

$$\frac{\sum_{j \neq i} A_R(k) s_t^j(k)}{\sum_{j \neq i} A_R(k)} = d_{t+1}(k) + \frac{1}{N-1} \sum_{j \neq i} \varepsilon_{jt}(k)$$

which is a random variable with mean 0 and variance $\frac{\tau_k^2}{N-1}$. Thus, the expected value for the dividend becomes:

$$E [d_{t+1}(k) | p_t(k), s_t^i] = \frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2} \sum_j s_t^j(k)$$

Substituting the guess for $p_t(k)$ into the market clearing condition gives:

$$\begin{aligned} N \left(\frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2} - B_R(k) \right) \mu_k + \sum_i \left(\frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2} - A_R(k) \right) s_t^i \\ = \alpha N \{ \text{Cov}_t(p_{t+1}(k), d_{t+2}(k) + p_{t+2}(k) - p_{t+1}(k)) + \text{Var}_t(d_{t+1}(k) + p_{t+1}(k)) \} \\ - \alpha N \{ \text{Cov}_t(p_{t+1}(k), d_{t+2}(k)) + \text{Var}_t(d_{t+1}(k)) \} = 0 \end{aligned}$$

Thus:

$$A_R(k) = \frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2}; B_R(k) = \frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2}$$

Hence, a REE exists. The average signal can be extracted from the price, which shows that it is fully revealing.

Under REE, all agents are identical in both preferences and information, and thus have the same demand function. Thus, in any REE there is no trade: $V_t(k) = 0$ for any k and period $0 < t < T$.

4.3 Dynamic Cursed Expectation Equilibrium

In full DCEE all agents use DCE. Period t agents learn from their own private information about d_{t+1} but fail to take into account the information in p_t when submitting their demand

curve. A strategic interpretation is that agents don't understand how the aggregate behavior of other agents relate to their information, when this action is in the future. When considering the continuum of agents of each type interpretation, the bias is present regarding *all* the other agents, out-group and in-group alike.

The price is given by:

$$p_t(k) = \frac{1}{N} \sum_i E [d_{t+1}(k) | s_t^i] + E [p_{t+1}(k)] - \alpha \{ \text{Cov} (p_{t+1}(k), d_{t+2}(k)) + \text{Var} (d_{t+1}(k) | s_t) \} \quad (10)$$

where p_{t+1}, d_{t+2} are treated as independent. The variance $\text{Var} (d_{t+1}(k) | s_t^i)$ does not depend on the agent: as s_t^i 's are drawn from the same distribution, the variance of $d_{t+1}(k)$ given a signal is the same for any such signal.

$$p_t(k) = \frac{1}{N} \sum_i E_t^i [d_{t+1}(k) | s_t^i] + E_t [p_{t+1}(k)] - \alpha \text{Var}_t (d_{t+1}(k) | s_t)$$

To show that DCEE exists in this application, guess a linear and symmetric DCEE solution, as in the case of REE.

$$p_t(k) = \sum A_{DC}(k) s_t^i(k) + B_{DC}(k) \mu_k + E_t [p_{t+1}(k)] - \alpha \text{Var}_t (d_{t+1}(k) | s_t)$$

Here, $E_t^i [d_{t+1}(k)] = \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} s_t^i$, and similarly to the REE solution, market clearing gives:

$$A_{DC}(k) = \frac{1}{N} \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2}; B_{DC}(k) = \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2}$$

Unlike REE, a DCEE typically exhibits different demand functions for different traders, as those depend on private information. Thus, trade is expected.

Except at the final trading date, risky assets have higher expected prices in DCEE than in REE. The bias shrinks as the end approaches and reverses sign at the final trading date, where cursed traders, each pricing off a single signal, demand more risk compensation than rational traders. The proof appears in [Appendix A](#).

Proposition 4.1. *For every $N \geq 2$ and every trading date $t = 0, \dots, T - 1$,*

$$E [p_t^{DCEE}(k) - p_t^{REE}(k)] = \frac{\alpha \sigma_k^4}{\sigma_k^2 + \tau_k^2} \left[(T - 1 - t) - \frac{(N - 1) \tau_k^2}{N \sigma_k^2 + \tau_k^2} \right].$$

In particular: (i) the bias strictly decreases over time; (ii) at the final trading date $t = T - 1$ it is negative and the asset is undervalued; (iii) whenever $\sigma_k \geq \tau_k$, the asset is overvalued at every $t \leq T - 2$, for every N ; (iv) wherever the bias is positive, it increases with σ_k^2 .

4.4 Partially Dynamic Cursed Expectations Equilibrium

Does 4.1 continue to hold when traders are only partially cursed? In χ -DCEE all agents use χ -DCE. An alternative approach could include a combination of agents that use full DCE and agents that use RE, but this approach would be prone to discontinuities due to the finite number of agents. With χ -DCEE, the price is given by:

$$p_t(k) = \frac{1}{N} \sum_i \{ \chi E[d_{t+1}(k) | s_i] + (1 - \chi) E[d_{t+1}(k) | p_t, s_t^i] \} + E_t[p_{t+1}(k)] \\ - \alpha \{ (1 - \chi) \text{Cov}_t^R(p_{t+1}(k), d_{t+2}(k)) + \chi \text{Var}_t(d_{t+1}(k) | s_t) + (1 - \chi) \text{Var}_t(d_{t+1}(k) | p_t) \} \quad (11)$$

for all t , where $E[p_{t+1}(k) | p_t, s_t^i] = E[p_{t+1}(k) | s_t^i] = E[p_{t+1}(k)]$, and $\text{Cov}_t^R(p_{t+1}(k), d_{t+2}(k))$ is the real ex-ante (as was used in REE).

First, guess a linear and symmetric solution, as before:

$$p_t(k) = \sum A_{\chi DC}(k) s_t^i(k) + B_{\chi DC}(k) \mu_k + E_t[p_{t+1}(k)] \\ - \alpha \{ (1 - \chi) \text{Cov}_t^R(p_{t+1}(k), d_{t+2}(k)) + \chi \text{Var}_t(d_{t+1}(k) | s_t) + (1 - \chi) \text{Var}_t(d_{t+1}(k) | p_t) \}$$

Recall:

$$E[d_{t+1}(k) | s_t^i] = \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} s_t^i \\ E[d_{t+1}(k) | p_t(k), s_t^i] = \frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2} \sum_j s_t^j(k)$$

Substituting the guess for $p_t(k)$ into the market clearing condition gives:

$$N \left\{ \chi \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} + (1 - \chi) \frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2} - B_{\chi DC}(k) \right\} \mu_k \\ + \left\{ \chi \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} + (1 - \chi) \frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2} N - A_{\chi DC}(k) N \right\} \sum_i s_t^i = 0$$

Hence,

$$A_{\chi DC}(k) = \chi \frac{1}{N} \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} + (1 - \chi) \frac{\sigma_k^2}{N\sigma_k^2 + \tau_k^2}; \quad B_{\chi DC}(k) = \chi \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} + (1 - \chi) \frac{\tau_k^2}{N\sigma_k^2 + \tau_k^2}$$

That is, precisely,

$$A_{\chi DC}(k) = \chi A_{DC}(k) + (1 - \chi) A_R(k); \quad B_{\chi DC}(k) = \chi B_{DC}(k) + (1 - \chi) B_R(k)$$

Thus, an equilibrium exists.

Noting that Equation 11 is simply a linear combination of Equations 9 and 10, the following conclusion is immediate.

Conclusion 4.2. *For every $N \geq 2$ and every trading date t , the expected price bias of χ -DCEE relative to REE equals χ times the bias of Proposition 4.1; in particular, all conclusions of Proposition 4.1 hold with the bias scaled by χ .*

4.5 Trade Volume

If the initial endowment is Pareto efficient, in REE there is no trade: traders have identical utilities, and they all use all of the relevant information. Thus, their demand functions are identical between traders and over time. In contrast, DCEE exhibits trade. Recall that demand is given by:

$$\begin{aligned} & x_{it+1}(k) \\ &= \frac{E_t^i [d_{t+1}(k) + p_{t+1}(k)] - p_t(k) - \alpha \text{Cov}_t^i(p_{t+1}(k), (d_{t+2}(k) + p_{t+2}(k) - p_{t+1}(k)) x_{it+2}(k))}{\alpha \text{Var}_t^i(d_{t+1}(k) + p_{t+1}(k))} \end{aligned} \quad (12)$$

Since all traders are identical, it suffices to examine the expected volume of trade generated by a single agent. The volume of trade in asset k generated by trader i is $|x_{it+2}(k) - x_{it+1}(k)|$, where ex-ante $x_{it+2}(k)$ and $x_{it+1}(k)$ are random variables that are independent from one another due to the absence of wealth effects.

The choice of agent i , $x_{it+1}(k)$, depends on the realization of the signal s_t^i . The constant part of 12 is identical across time-periods, thus disappears in the subtraction $|x_{it+2}(k) - x_{it+1}(k)|$. This is because the effect of future price and demand on the covariance is time independent (the time dependent part has no uncertainty). The non-constant part is:

$$\frac{E_t^i [d_{t+1}(k) + p_{t+1}(k)] - p_t(k)}{\alpha \text{Var}_t^i(d_{t+1}(k) + p_{t+1}(k))}$$

Where the denominator, $\alpha \text{Var}_t^i(d_{t+1}(k) + p_{t+1}(k))$, is constant over time for all $t < T - 1$. In the last period, $p_T = 0$ thus $\text{Var}_{T-1}^i(d_T(k) + p_T(k)) = \text{Var}_{T-1}^i(d_T(k))$.

The following result characterizes the volume of trade and is proven in Appendix A.

Proposition 4.3. *Let $0 < t < T - 3$. The expected per-agent volume of trade in asset k is*

$$E_{<t} [v_t(k)] = \frac{2\tau_k (\sigma_k^2 + \tau_k^2) \sqrt{N(N-1)/\pi}}{\alpha [N(\sigma_k^4 + \sigma_k^2 \tau_k^2 + \tau_k^4) + \sigma_k^2 \tau_k^2]}.$$

It is strictly positive, strictly decreasing in σ_k , and strictly increasing in N . When uncertainty about the dividend is higher, demand is scaled down in absolute value and so is the volume. When more traders are present, there is more disparity of opinion. As the number of trader approaches infinity,

$$\lim_{N \rightarrow \infty} E_{<t} [v_t(k)] = \frac{2\tau_k (\sigma_k^2 + \tau_k^2)}{\alpha \sqrt{\pi} (\sigma_k^4 + \sigma_k^2 \tau_k^2 + \tau_k^4)}.$$

Moreover, trade is driven entirely by the dispersion of private noise: common news moves all traders' expectations, demands, and the price in parallel, and cancels from relative positions. Accordingly, the expected volume is hump-shaped in τ_k : with no noise there is no disagreement and no trade, while with very noisy signals demand becomes unresponsive to information and trade dies out again.

5 Conclusion

In this paper, I define Dynamic Cursed Expectations and Dynamic Cursed Expectations Equilibrium, and show that DCEE exists in financial economies with imminent private information in which traders have CARA preferences and access to riskless storage. I study an asset-pricing application where agents have mean-variance preferences, the assets are all in positive supply, the dividend for each asset is normal and IID across time periods, and agents receive private information about next period's dividends. I find that assets are overpriced in DCEE relative to REE. DCEE traders fail to learn hypothetically from prices: they neglect the correlation between each future price and the following period's dividend. They consider different channels to profit from their present choice of quantity: one through reselling the asset, and the other through a dividend pay in a later period. Thinking that those quantities are uncorrelated, traders underestimate the variance of future revenues and overvalue the risky assets. In addition, DCEE generates trade, where in REE there is no trade.

The application shows that in DCEE, risk aversion can lead to excess risk taking. This outcome is expected more general whenever "good things go together". DCE agents have the correct estimation of future expected payoffs based on past information. However, when they are risk averse, their utility also depends on the variance of expected payoffs. When this variance depends on the covariance between future sources of income at different time period, it will be distorted.

DCE and DCEE are defined in a general setting and can be applied to a variety of models in macroeconomics and finance. Contingent thinking can be important when people take a mortgage to buy a house, when firms make irreversible investment decisions, or when policy makers make forecasts on the future of inflation and macroeconomic shocks. While DCEE

was defined for financial economies, it can be extended to other settings: in DCEE, all agents behave according to DCE, and the appropriate markets clear. The study of the implications of DCEE in those applications and in others is left to future work.

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A Proofs omitted from the main text

A.1 Proof of Theorem 3.6

The proof constructs the equilibrium by backward induction on the trading date. The induction hypothesis at date t is:

(IH _{$t+1$}) Measurable pricing functions $\varrho_r : \hat{\Omega} \rightarrow [(T-r)\underline{d}, (T-r)\bar{d}]^K$ for $r = t+1, \dots, T-1$ have been constructed (with $\varrho_T \equiv 0$), such that period- r prices $p_r = \varrho_r(\hat{\omega}_r)$ depend only on the period- r signal profile.

Fix t and assume (IH _{$t+1$}). Step 4 constructs the present pricing function ϱ_t and shows (IH _{t}). All integrability claims below are immediate because dividends and (under the induction hypothesis) all future prices take values in fixed compact sets, so every integrand appearing in the proof is bounded on the relevant domain.

Step 1: Beliefs are price-free

Lemma A.1. *Let $\hat{\eta}_i^t = (d^t, p^{t-1}, \omega_i^t)$ be a morning history consistent with ρ . For almost every such history, the posterior distribution of the future exogenous variables $(d_{t+1}, \dots, d_T; \hat{\omega}_{t+1}, \dots, \hat{\omega}_T)$ given $\hat{\eta}_i^t$ coincides with the conditional distribution given ω_{it} alone, denoted $\lambda_{it}(\cdot \mid \omega_{it})$.*

Proof. Under (IH _{$t+1$}) and Assumption 3.4, every variable in $\hat{\eta}_i^t = (d^t, p^{t-1}, \omega_i^t)$ is a function of $(d^t, \varepsilon^{t-1}, \varepsilon_{it})$. Past prices are functions of (d^t, ε^{t-1}) : for $r \leq t-1$, $p_r = \varrho_r(\hat{\omega}_r)$ with $\hat{\omega}_r = g(d_{r+1}, \varepsilon_r)$, where $g = (g_i)_i$. The signals vector for player i , ω_i^t , is a function of $(d^{t+1}, \varepsilon_i^t)$. Note ω_{it} is the only conditioning variable involving d_{t+1} . Because dividends are i.i.d. and the noise array is independent of them and i.i.d., the vector $(d^t, \varepsilon^{t-1}, \varepsilon_i^{t-1})$ is independent of $(d_{t+1}, d_{t+2}, \dots; \varepsilon_t^{-i}, \varepsilon_{t+1}, \dots)$. Hence, conditioning on the full morning history refines conditioning on ω_{it} only by variables independent of the future, and the two posteriors agree almost surely. The description of the marginals follows: d_{t+1} is informed by ω_{it} ; p_{t+1} is a function of $(d_{t+2}, \varepsilon_{t+1})$, which is independent of (d_{t+1}, ω_{it}) , so within the period- $(t+1)$ triple the pair (d_{t+1}, p_{t+1}) is independent. For $r \geq t+2$ nothing in the period- r triple is correlated with ω_{it} . Adopting $\lambda_{it}(\cdot \mid \omega_{it})$ as the belief at every morning history makes beliefs price-free identically, not merely almost surely. $\square$

As a result, the DCE per-period marginals of (d_r, p_r, ω_{ir}) , $r > t$, are, for a canonical distribution adopted at every morning history (on- or off-path):

1. for $r = t+1$: $d_{t+1} \sim \lambda$ -posterior given ω_{it} , and independently $p_{t+1} = \varrho_{t+1}(\hat{\omega}_{t+1})$ with $\hat{\omega}_{t+1} = g(d_{t+2}, \varepsilon_{t+1})$ where $g = (g_i)_i$;
2. for $r \geq t+2$: the unconditional distribution of $(d_r, \varrho_r(\hat{\omega}_r), \omega_{ir})$.

Step 2: The planning problem separates, demand is wealth-free

Fix a signal realization ω_{it} . Let m_{ir} , $r = t + 1, \dots, T$, denote the period- r marginals of Lemma A.1, and let $\mathbb{M}_{it} = \bigotimes_{r>t} m_{ir}$ be the DCE product belief. A morning plan prescribes a demand curve (c, b', x') as a function of the current price p_t , and a plan for each future period r . The DCE expected utility of a plan is the integral of U_i under $\mathbb{M}_{it}$, as in Equation 2.

Lemma A.2. *For all $r = t, \dots, T$, define $\kappa_{ir} = \gamma_i / (T - r + 1)$. Define recursively, backward from $r = T$, the functions $\Psi_{ir} : \mathbb{R}^K \rightarrow \mathbb{R}$, $A_{ir} : P \rightarrow \mathbb{R}$, and $M_{ir} : P \rightarrow \mathbb{R}$:*

$$\begin{aligned} \Psi_{ir}(x) &= \int A_{ir+1}(\tilde{p}) e^{-\kappa_{ir+1}(\tilde{d} + \tilde{p}) \cdot x} dm_{ir+1}(\tilde{d}, \tilde{p}, \tilde{\omega}), \quad x \in \mathbb{R}^K, \quad r \leq T - 1, \quad (13) \\ A_{iT}(\cdot) &\equiv \delta_{iT}, \text{ and for } r \leq T - 1, A_{ir}(\tilde{p}) = C_{ir} \cdot M_{ir}(\tilde{p})^{\kappa_{ir}/\kappa_{ir+1}} \\ M_{ir}(\tilde{p}) &= \min_{x \in \mathbb{R}^K} e^{\kappa_{ir+1}\tilde{p} \cdot x} \Psi_{ir}(x) \end{aligned}$$

where C_{ir} is a positive constant that depends only on $(\gamma_i, \delta_{ir}, \kappa_{ir+1})$. Here and throughout, $(\tilde{d}, \tilde{p}, \tilde{\omega}) \sim m_{ir+1}$ denotes the period- $(r + 1)$ dividend, price, and signal under the trader's date- t belief. Essentially, $\tilde{d} + \tilde{p}$ is the payoff she perceives on a unit of the assets carried into period $r + 1$. Then:

1. for every current price $p_t \in B_{it}$ (defined in claim 3 below), the planning problem at the morning history has a solution, and its value equals $-A_{it}(p_t) e^{-\kappa_{it}W}$, where W is wealth;
2. the optimal period- t risky demand is

$$\varphi_{it}(p_t) = \operatorname{argmin}_{x \in \mathbb{R}^K} e^{\kappa_{it+1} p_t \cdot x} \Psi_{it}(x), \quad (14)$$

which is independent of the wealth W_{it} , hence of the trader's entering holdings, storage, and trading history, and depends on the state only through ω_{it} ;

3. the minimum in (14) and in the definition of M_{ir} is attained and unique whenever the price lies in B_{ir} , defined as the interior of the convex hull of the support of the perceived payoff $\tilde{d} + \tilde{p}$. This is an open convex subset of the box $((T - r)\underline{d}, (T - r)\bar{d})^K$ that contains every price the construction produces.

Proof. The argument is by backward induction on r within the planning problem, using dynamic programming under $\mathbb{M}_{it}$: because $\mathbb{M}_{it}$ is a product measure, the conditional distri-

bution of $(\tilde{d}, \tilde{p}, \tilde{\omega})$ is m_{ir+1} , regardless of the path until r . Hence, it suffices to show that the value functions have the above exponential form and that measurable optimal policies exist.

For $r = t, \dots, T$, define value functions $V_{ir}(W, q)$ backward, where W is wealth and q its current price (for $r = t$, $q = p_t$ is the argument of the demand curve; for $r > t$, q is the price $\tilde{p}_r$). At $r = T$ the choice is trivial—the trader consumes her wealth. Set:

$$V_{iT}(W, q) = -\delta_{iT}e^{-\gamma_i W} = -A_{iT}e^{-\kappa_{iT}W}$$

(recall $\kappa_{iT} = \gamma_i$ and $A_{iT} \equiv \delta_{iT}$). For $r < T$, a trader with wealth W given price q chooses consumption c and positions x' , with storage $b' = W - c - q \cdot x'$ determined by the budget identity, so that next period's wealth is $W' = (W - c) + (\tilde{d} + \tilde{p} - q) \cdot x'$. For $r = t, \dots, T - 1$, the value function is:

$$V_{ir}(W, q) = \sup_{c \in \mathbb{R}, x' \in \mathbb{R}^K} \left\{ -\delta_{ir}e^{-\gamma_i c} + \int V_{ir+1}((W - c) + (\tilde{d} + \tilde{p} - q) \cdot x', \tilde{p}) dm_{ir+1}(\tilde{d}, \tilde{p}, \tilde{\omega}) \right\}. \quad (15)$$

This is the Bellman equation of the planning problem: because $\mathbb{M}_{it}$ is a product measure, the continuation belief at every node is $\bigotimes_{s>r} m_{is}$ regardless of the path up to r , so (W, q) is the only payoff-relevant state. Suppose inductively that $V_{ir+1}(W', \tilde{p}) = -A_{ir+1}(\tilde{p})e^{-\kappa_{ir+1}W'}$ with A_{ir+1} positive and measurable. Substituting into (15) and factorizing the exponential, $e^{-\kappa_{ir+1}W'} = e^{-\kappa_{ir+1}(W-c)} e^{\kappa_{ir+1}q \cdot x'} e^{-\kappa_{ir+1}(\tilde{d}+\tilde{p}) \cdot x'}$, the integral collapses by (13), and

$$V_{ir}(W, q) = \sup_{c, x'} \left\{ -\delta_{ir}e^{-\gamma_i c} - e^{-\kappa_{ir+1}(W-c)} e^{\kappa_{ir+1}q \cdot x'} \Psi_{ir}(x') \right\}. \quad (16)$$

The choice variable x' enters (16) only through the factor $e^{\kappa_{ir+1}q \cdot x'} \Psi_{ir}(x') > 0$, which multiplies a term that does not involve x' . Hence the optimal choice minimizes that factor, independently of W and c . Substituting the minimized value $M_{ir}(q)$ into (16) leaves a one-dimensional consumption–savings problem,

$$V_{ir}(W, q) = \sup_{c \in \mathbb{R}} \left\{ -\delta_{ir}e^{-\gamma_i c} - M_{ir}(q) e^{-\kappa_{ir+1}(W-c)} \right\}, \quad (17)$$

which is smooth, strictly concave, and tends to $-\infty$ as $|c| \rightarrow \infty$. The maximizer c^* therefore exists, is unique, and is characterized by the first-order condition

$$\gamma_i \delta_{ir} e^{-\gamma_i c^*} = \kappa_{ir+1} M_{ir}(q) e^{-\kappa_{ir+1}(W-c^*)} : \quad (18)$$

the marginal utility of consuming today equals the marginal value of wealth carried forward. The value then has a closed form. Write λ for the common value of the two sides of (18).

The two terms of (17) at the optimum equal $-\lambda/\gamma_i$ and $-\lambda/\kappa_{ir+1}$, so

$$V_{ir}(W, q) = -\lambda \left(\frac{1}{\gamma_i} + \frac{1}{\kappa_{ir+1}} \right) = -\frac{\lambda}{\kappa_{ir}}, \quad \text{where} \quad \frac{1}{\kappa_{ir}} = \frac{1}{\gamma_i} + \frac{1}{\kappa_{ir+1}}.$$

This is the harmonic recursion for κ : risk tolerances add across the periods over which a wealth shock will be smoothed. Starting from $\kappa_{iT} = \gamma_i$, it gives $\kappa_{ir} = \gamma_i / (T - r + 1)$. It remains to express λ in terms of (W, q) : raising the two expressions for λ in (18) to the powers $1/\gamma_i$ and $1/\kappa_{ir+1}$ respectively and multiplying them,

$$\lambda^{1/\kappa_{ir}} = (\gamma_i \delta_{ir})^{1/\gamma_i} (\kappa_{ir+1} M_{ir}(q))^{1/\kappa_{ir+1}} e^{-W},$$

so that $V_{ir}(W, q) = -A_{ir}(q) e^{-\kappa_{ir}W}$ with

$$A_{ir}(q) = \underbrace{\frac{1}{\kappa_{ir}} (\gamma_i \delta_{ir})^{\kappa_{ir}/\gamma_i} \kappa_{ir+1}^{\kappa_{ir}/\kappa_{ir+1}}}_{=C_{ir}} \cdot M_{ir}(q)^{\kappa_{ir}/\kappa_{ir+1}},$$

A_{ir} is strictly positive, and its measurability in q is inherited from that of M_{ir} . This closes the induction on the value functions.

For the plan itself: at every node, the optimal position is the unique minimizer of claim 3 and the optimal consumption is the unique solution of (18), both measurable functions of (W, q) (measurable maximum theorem, with uniqueness making the selection unambiguous), so the recursion's policies assemble into a well-defined plan. Because the horizon is finite and all integrands above are bounded on the compact dividend and price ranges, the plan attains $V_{it}(W, p_t)$, and no plan attains more. This proves claim 1, and claim 2 follows because the period- t position was chosen in (16) to minimize $e^{\kappa_{it+1}p_t \cdot x'} \Psi_{it}(x')$, before W or c entered the problem.

It remains to prove claim 3, which also establishes that the objects used above are well defined. Write $\kappa = \kappa_{ir+1}$ and

$$e^{\kappa q \cdot x} \Psi_{ir}(x) = \exp \{ \kappa q \cdot x + \log \Psi_{ir}(x) \}.$$

The argument establishes three properties: $\log \Psi_{ir}$ is finite and smooth, strictly convex, and for $q \in B_{ir}$ the exponent tends to ∞ at $\|x\| \rightarrow \infty$. Such a function on $\mathbb{R}^K$ attains a unique minimum, so the three properties together prove the claim.

Finiteness and smoothness. For each x , the integrand in (13) is bounded: $\tilde{d} + \tilde{p}$ ranges in a compact set, and A_{ir+1} is bounded above uniformly, since $M_{ir+1}(\tilde{p}) \leq \Psi_{ir+1}(0) < \infty$ for every $\tilde{p}$. Moreover, $A_{ir+1} > 0$ at every price the construction produces: such prices lie in

B_{ir+1} , where the minimum defining M_{ir+1} is attained and strictly positive. This is due to the induction hypothesis, with the trivial base case $A_{iT} \equiv \delta_{iT}$. Hence Ψ_{ir} is finite, and by dominated convergence it is smooth, with derivatives obtained by differentiating under the integral sign.

Strict convexity. The Hessian is

$$\nabla^2 \log \Psi_{ir}(x) = \kappa^2 \text{Cov}_{Q_x}(\tilde{d} + \tilde{p}),$$

where Q_x is the probability measure with density proportional to $A_{ir+1}(\tilde{p}) e^{-\kappa(\tilde{d} + \tilde{p}) \cdot x}$ with respect to m_{ir+1} . That density is strictly positive, so Q_x and m_{ir+1} share the same null sets. Under m_{ir+1} the pair $(\tilde{d}, \tilde{p})$ is independent (Lemma A.1), and $\tilde{d}$ has a posterior distribution whose density with respect to ν is proportional to the strictly positive likelihood (Assumption 3.4). The posterior therefore has exactly the support of ν , which no hyperplane contains (Assumption 3.3). Consequently, for every $v \neq 0$, the random variable $v \cdot \tilde{d}$ is nonconstant under Q_x , and by independence so does $v \cdot (\tilde{d} + \tilde{p})$. The Hessian is therefore positive definite, and $\log \Psi_{ir}$ is strictly convex.

Coercivity. First,

$$\nabla \log \Psi_{ir}(x) = -\kappa \mathbb{E}_{Q_x}[\tilde{d} + \tilde{p}],$$

and $\tilde{d} + \tilde{p}$ takes values in $[(T-r)\underline{d}, (T-r)\bar{d}]^K$. Fix $q \in B_{ir}$ and a unit vector u , and consider the function $h(\tau) = \kappa q \cdot (\tau u) + \log \Psi_{ir}(\tau u)$, which is convex and

$$h'(\tau) = \kappa u \cdot \left(q - \mathbb{E}_{Q_{\tau u}}[\tilde{d} + \tilde{p}] \right).$$

As $\tau \rightarrow \infty$, the tilted measure $Q_{\tau u}$ concentrates where $u \cdot (\tilde{d} + \tilde{p})$ is smallest. Write m_u for the infimum of $u \cdot (\tilde{d} + \tilde{p})$; then, the event $\left\{ u \cdot (\tilde{d} + \tilde{p}) \leq m_u + \varepsilon/2 \right\}$ has positive mass for every $\varepsilon > 0$ and outweighs its complement exponentially as $\tau \rightarrow \infty$, so $Q_{\tau u}(u \cdot (\tilde{d} + \tilde{p}) \geq m_u + \varepsilon) \rightarrow 0$ and $\liminf_{\tau} h'(\tau) \geq \kappa(u \cdot q - m_u)$. Because q lies in the interior of the convex hull of the support, $u \cdot q > m_u$ for every unit vector u ; and since $u \mapsto m_u$ is continuous (up to sign, the support function of a compact set), $\theta_q := \min_{\|u\|=1} (u \cdot q - m_u) > 0$. Along every ray, h' is therefore eventually at least $\kappa\theta_q/2$. It then follows that $e^{\kappa q \cdot x} \Psi_{ir}(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$. Finally, Step 4 below evaluates M_{ir} and the demand (14) only at prices in the relevant sets B_{ir} , where the claim applies. □

Step 3: The market as a planner problem

Fix a signal profile $\hat{\omega}_t = (\omega_{it})_i$. Define, for each trader,

$$f_{it}(x) = -\frac{1}{\kappa_{it+1}} \log \Psi_{it}(x), \quad x \in \mathbb{R}^K.$$

By Lemma A.2, f_{it} is smooth and strictly concave with $\nabla f_{it}(x) = \mathbb{E}_{Q_x^i} [\tilde{d} + \tilde{p}]$, and the demand curve (14) is characterized by the first-order condition $p_t = \nabla f_{it}(x)$. Economically, $f_{it}(x)$ is trader i 's risk-adjusted valuation of carrying the position x into period $t + 1$ (a certainty-equivalent-type object under her DCE belief, with states weighted by their continuation values), and the first-order condition says she trades in each asset until its price equals her marginal valuation.

The following lemma establishes existence and uniqueness of the market-clearing price by exhibiting it as the shadow price of a planner problem: the problem of allocating the aggregate supply S across traders to maximize $\sum_i f_{it}(x_i)$. At the planner's optimum, all traders' marginal valuations ∇f_{it} are equated to a common multiplier. That multiplier is the price at which each trader's demand is optimal and the market clears.

Lemma A.3. *The problem*

$$\max \sum_{i \in I} f_{it}(x_i) \quad \text{subject to} \quad \sum_{i \in I} x_i = S \quad (19)$$

has a unique solution $(x_{it+1}^*)_i$, and its Lagrange multiplier $p_t^* = \nabla f_{it}(x_{it+1}^*)$ (for any i) is the unique market-clearing price: $\sum_i \varphi_{it}(p_t^*) = S$. Moreover p_t^* lies in B_{it} for every trader i and is a continuous (hence measurable) function of $\hat{\omega}_t$ only.

Proof. Throughout the proof, let Σ denote the support of the perceived payoff $\tilde{d} + \tilde{p}$ under the traders' beliefs m_{it+1} , $i \in I$. The support is common to all traders: by Assumption 3.4 and Lemma A.1, the positive likelihood gives the posterior of $\tilde{d}$ the same support as ν for every trader and every signal, while $\tilde{p}$ has the same unconditional distribution for every i . By independence, the joint support is the closure of the sum of the two supports. The support Σ is compact, and by Assumption 3.3 it is not contained in any affine hyperplane.

The main step is to show that the planner's objective tends to $-\infty$ as $\|(x_i)_{i \in I}\| \rightarrow \infty$ within the constraint set, existence of a maximizer then follows.

Formally, fix $\varepsilon > 0$. For any $z_0 \in \Sigma$, keeping in (13) a mass of perceived payoffs $\tilde{d} + \tilde{p}$ that are within distance ε of z_0 gives

$$\Psi_{it}(x) \geq \pi_\varepsilon e^{-\kappa_{it+1}(z_0 \cdot x + \varepsilon \|x\|)},$$

where $\pi_\varepsilon > 0$ is a lower bound on the A_{it+1} -weighted mass of the ε -ball around a support point. The bound is assumed to be uniform over $z_0 \in \Sigma$; a uniform bound can be constructed using a finite cover of Σ with $\varepsilon/2$ -balls, and taking the minimum over finitely many centers. Taking logarithms, dividing by $-\kappa_{it+1}$, and minimizing over $z_0 \in \Sigma$,

$$f_{it}(x) \leq \min_{z \in \Sigma} z \cdot x + \varepsilon \|x\| + C_\varepsilon, \quad (20)$$

for a finite constant C_ε (depending on the state).

Now take any sequence of allocations on the constraint set with $\|(x_i)_i\| \rightarrow \infty$, and pass to a subsequence along which the directions $\left(x_i / \|(x_j)_j\|\right)_i$ converge to some $(v_i)_i$ with $\sum_i v_i = 0$ and $(v_i)_i \neq 0$: a pure reallocation. Taking each player's worst-case value over Σ and summing those over all players results in a strictly negative number. For any j with $v_j \neq 0$, $z \cdot v_j$ is nonconstant on Σ (since Σ is not contained in a hyperplane), so there is $z^* \in \Sigma$ with $\min_{z \in \Sigma} z \cdot v_j < z^* \cdot v_j$; hence

$$\sum_i \min_{z \in \Sigma} z \cdot v_i \leq \min_{z \in \Sigma} z \cdot v_j + \sum_{i \neq j} z^* \cdot v_i = \min_{z \in \Sigma} z \cdot v_j - z^* \cdot v_j < 0.$$

The map $(v_i)_i \mapsto \sum_i \min_{z \in \Sigma} z \cdot v_i$ is continuous, so its supremum over the compact set of unit-norm directions with $\sum_i v_i = 0$ equals some $-2\eta < 0$. Since each $\|v_i\| \leq 1$, choosing $\varepsilon = \eta/N$ in (20) and summing over traders leads to $\sum_i f_{it}(x_i) \rightarrow -\infty$ along every sequence in the constraint set that tends to infinity.

The planner problem maximizes a continuous, strictly concave objective over a nonempty closed affine set on which it tends to $-\infty$ at infinity, so a unique maximizer $(x_{it+1}^*)_i$ exists. Market clearing is now the convex duality. The functions f_{it} are differentiable and the constraint is affine, so the Lagrange condition holds exactly: there is $p_t^* \in \mathbb{R}^K$ with $\nabla f_{it}(x_{it+1}^*) = p_t^*$ for every i . By Lemma A.2, this first-order condition characterizes the unique demand at p_t^* : $x_{it+1}^* = \varphi_{it}(p_t^*)$, hence $\sum_i \varphi_{it}(p_t^*) = S$ and markets clear at p_t^* . Conversely, if some price p' clears the market, the demands $(\varphi_{it}(p'))_i$ satisfy the common-gradient condition, hence solve (19) by concavity and coincide with $(x_{it+1}^*)_i$ by uniqueness. Thus $p' = \nabla f_{it}(x_{it+1}^*) = p_t^*$; The clearing price is unique.

It remains to show that p_t^* lies in B_{it} and that it is continuous in the signal profile. For containment, $p_t^* = \nabla f_{it}(x_{it+1}^*) = E_{Q_{x^*}^i} [\tilde{d} + \tilde{p}]$ holds simultaneously for every trader i . As $\tilde{d} + \tilde{p}$ is nonconstant in every direction, the expectation necessarily lies in the interior of the convex hull of its support: $p_t^* \in B_{it}$ for every i .

For continuity, note first that the posterior of d_{t+1} is continuous in the signal: this is by Bayes' rule, continuity of the likelihood, and compactness of D . Consequently Ψ_{it} , and with

it f_{it} and ∇f_{it} , are jointly continuous in (x, ω_{it}) : their integrands are bounded continuous functions of $(\tilde{d}, x)$ after integrating out $\tilde{p}$ against its fixed distribution, and the remaining integral is against the continuously varying posterior. The bound in (20) holds locally uniformly in the signal: taking the ball in the definition of π_ε to be open, the weighted mass of an open set is lower semicontinuous along a weakly continuous family of posteriors, so π_ε is locally bounded away from zero, C_ε is locally bounded, and the planner's maximizers remain in one fixed compact set for all signal profiles near any given one. On that compact set the maximum theorem applies as the objective is jointly continuous and the maximizer is unique. Hence $(x_{it+1}^*)_i$ and $p_t^* = \nabla f_{1t}(x_{1t+1}^*)$ are continuous in $\hat{\omega}_t$. Since each f_{it} depends on the state only through ω_{it} (Lemma A.2), the planner problem depends on it only through $\hat{\omega}_t$, and we may define $\varrho_t(\hat{\omega}_t) = p_t^*$. □

Step 4: The backward induction

The pieces now assemble into the pricing function. At $t = T - 1$, (IH_T) holds trivially ($\varrho_T \equiv 0$, consistent with $p_T = 0$). Given (IH_{t+1}) , Lemmas A.1–A.3 deliver a continuous function $\varrho_t : \hat{\Omega} \rightarrow [(T - t)\underline{d}, (T - t)\bar{d}]^K$ depending on the period- t signals only, with $\varrho_t(\hat{\omega}_t) \in \bigcap_i B_{it}$ at every state. This establishes (IH_t) . Descending from $t = T - 1$ to $t = 1$ produces the full pricing function $\rho = (\varrho_t)_{t=1}^{T-1}$.

Step 5: From the construction to a DCEE

It remains to exhibit the equilibrium objects: a plan for every trader at every morning history that satisfies the two conditions of the definition of DCEE.

Plans. At each morning history $\hat{\eta}_i^t$ (on- or off-path) with entering state (b_{it}, x_{it}) implied by the trader's own earlier plans and the recorded prices, attach the morning plan constructed in Lemma A.2. In period- t it maps each current price $p_t \in B_{it}$ to the budget-balancing triple $(c_{it}, b_{it+1}, x_{it+1})$ with $x_{it+1} = \varphi_{it}(p_t)$ and (c_{it}, b_{it+1}) the optimal consumption/storage choice of Lemma A.2 at wealth W_{it} . For prices outside B_{it} , the triple is arbitrary as no optimal choice exists, its future components are the measurable optimal policies of the planning problem.

Condition 1 (optimality). At every morning history, the attached plan maximizes the DCE expected utility of Equation 2 at every current price $p_t \in B_{it}$, with respect to the canonical posterior $\lambda_{it}(\cdot \mid \omega_{it})$: this is Lemma A.2 (dynamic-programming verification). No price outside the closure of B_{it} admits an optimal choice at all: there the trader perceives an arbitrage, and the supremum of her expected utility is unattained. The plan is optimal at every price at which optimality is achievable, according with the convention stated with the

definition of DCEE in Section 3, and the market only ever clears at prices in $\bigcap_i B_{it}$ (Lemma A.3). By Lemma A.1, λ_{it} is a version of the Bayes posterior $b_{\rho,i}[\hat{\eta}_i^t]$ at almost every ρ -consistent history, and it is the adopted belief everywhere else; hence $\pi_i[\hat{\eta}_i^t] \in o_i^{DC}(x_i^t, b_{\rho,i}[\hat{\eta}_i^t])$ at every morning history.

Condition 2 (market clearing). Fix any ρ -consistent history at date t , generated by $(d^t, \hat{\omega}^t)$, with $p_t = \varrho_t(\hat{\omega}_t)$. Each trader's period- t demand at p_t is $\varphi_{it}(p_t) = x_{it+1}^*$ by Lemma A.3, and $\sum_i x_{it+1}^* = S = \sum_i e_i$: the risky-asset markets clear. Market clearing holds regardless of the traders' entering holdings and storage, so the induction hypothesis used in Lemma A.3 (aggregate entering holdings equal to S on the ρ -consistent path) is needed only for the economy's aggregate consistency, and holds there by this very construction at date $t - 1$.

The pair $(\rho, (\pi_i[\hat{\eta}_i^t])_{i, \hat{\eta}_i^t})$ therefore satisfies both conditions of the definition of DCEE, completing the proof of Theorem 3.6. $\blacksquare$

A.2 Proof of Proposition 4.1

First, note that $E[\frac{1}{N} \sum_i E[d_{t+1}(k) | s_t^i]] = E[E[d_{t+1}(k) | p_t^{REE}]] = \mu$, by the law of iterated expectations. Next, define

$$\Delta_t p_t^{REE}(k) = p_t^{REE}(k) - E_t[p_{t+1}^{REE}(k)]$$

and

$$\Delta_t p_t^{DCEE}(k) = p_t^{DCEE}(k) - E_t[p_{t+1}^{DCEE}(k)]$$

Then, for $t \leq T - 2$ (so that the covariance term is present):

$$\begin{aligned} E[\Delta_t p_t^{DCEE}(k) - \Delta_t p_t^{REE}(k)] \\ = -\alpha \text{Var}(d_{t+1}(k) | s_t) + \alpha \{ \text{Cov}(p_{t+1}^{REE}(k), d_{t+2}(k)) + \text{Var}(d_{t+1}(k) | p_t^{REE}) \} \end{aligned} \quad (21)$$

Since p_{t+1}^{REE} is a fully revealing, invertible function of $\sum_i s_{t+1}^i(k)$,

$$\text{Cov}(p_{t+1}^{REE}(k), d_{t+2}(k)) = \text{Cov}(E[d_{t+2}(k) | p_{t+1}^{REE}], d_{t+2}(k)) = \text{Var}(E[d_{t+2}(k) | p_{t+1}^{REE}]),$$

and by the law of total variance, together with the stationarity of the environment,

$$\text{Cov}(p_{t+1}^{REE}(k), d_{t+2}(k)) + \text{Var}(d_{t+1}(k) | p_t^{REE}) = \text{Var}(d(k)) = \sigma_k^2.$$

Substituting into Equation 21: for every N and every $t \leq T - 2$,

$$E [\Delta_t p_t^{DCEE}(k) - \Delta_t p_t^{REE}(k)] = \alpha (\sigma_k^2 - \text{Var}(d_{t+1}(k) | s_t^i)) = \frac{\alpha \sigma_k^4}{\sigma_k^2 + \tau_k^2} > 0.$$

At the final trading date $t = T - 1$ there is no continuation ($p_T = 0$), so the covariance term is absent and only the variance terms remain:

$$\begin{aligned} E [\Delta_{T-1} p_{T-1}^{DCEE}(k) - \Delta_{T-1} p_{T-1}^{REE}(k)] &= \alpha \{ \text{Var}(d_T(k) | p_{T-1}^{REE}) - \text{Var}(d_T(k) | s_{T-1}^i) \} \\ &= -\frac{\alpha(N-1)\sigma_k^4\tau_k^2}{(\sigma_k^2 + \tau_k^2)(N\sigma_k^2 + \tau_k^2)} < 0, \end{aligned}$$

using $\text{Var}(d | p^{REE}) = \frac{\sigma_k^2\tau_k^2}{N\sigma_k^2 + \tau_k^2}$ and $\text{Var}(d | s^i) = \frac{\sigma_k^2\tau_k^2}{\sigma_k^2 + \tau_k^2}$. Write $p_t = \Delta_t p_t + E_t[p_{t+1}]$ and iterate forward to $p_T = 0$. Since dividends and signals are i.i.d., each increment has the unconditional expectation computed above, so

$$\begin{aligned} E [p_t^{DCEE}(k) - p_t^{REE}(k)] &= \sum_{r=t}^{T-1} E [\Delta_r p_r^{DCEE}(k) - \Delta_r p_r^{REE}(k)] \\ &= \frac{\alpha \sigma_k^4}{\sigma_k^2 + \tau_k^2} \left[(T-1-t) - \frac{(N-1)\tau_k^2}{N\sigma_k^2 + \tau_k^2} \right]. \end{aligned}$$

Claims (i) and (ii) read off the formula. For (iii), the bias is positive at t if and only if $(T-1-t)(N\sigma_k^2 + \tau_k^2) > (N-1)\tau_k^2$. When $\sigma_k \geq \tau_k$, $\frac{(N-1)\tau_k^2}{N\sigma_k^2 + \tau_k^2} < 1 \leq T-1-t$ for every $t \leq T-2$ and every N , since it is equivalent to $N(\sigma_k^2 - \tau_k^2) + 2\tau_k^2 > 0$. For (iv), write the bias as $F(\sigma_k^2)[(T-1-t) - h(\sigma_k^2)]$ with $F(s) = \frac{\alpha s^2}{s + \tau_k^2}$ strictly increasing and $h(s) = \frac{(N-1)\tau_k^2}{Ns + \tau_k^2}$ strictly decreasing in s . Wherever the bracket is positive, both factors strictly increase in σ_k^2 .

A.3 Proof of Proposition 4.3

The volume of trade is

$$v_t(k) = \left| \frac{E_{t+1}^i [d_{t+2}(k) + p_{t+2}(k)] - p_{t+1}(k)}{\alpha \text{Var}_{t+1}^i (d_{t+2}(k) + p_{t+2}(k))} - \frac{E_t^i [d_{t+1}(k) + p_{t+1}(k)] - p_t(k)}{\alpha \text{Var}_t^i (d_{t+1}(k) + p_{t+1}(k))} \right|$$

Recall:

$$p_t(k) = \sum_i A_{DC}(k) s_t^i(k) + B_{DC}(k) \mu_k + E_t[p_{t+1}(k)] - \alpha \text{Var}_t(d_{t+1}(k) | s_t)$$

The denominators are identical for the two dates. Under DCE the trader conditions

on her own signal, so her perceived dividend variance is $\text{Var}(d_{t+1}(k) | s_t^i) = \frac{\sigma_k^2 \tau_k^2}{\sigma_k^2 + \tau_k^2} \equiv V_1(k)$. Since DCE preserves within-period joint distributions, her perceived variance of $p_{t+1}(k)$ is the true one, where the period- $(t+1)$ signals are correlated across traders through the common dividend $d_{t+2}(k)$: $\text{Var}\left(\sum_j s_{t+1}^j(k)\right) = N^2 \sigma_k^2 + N \tau_k^2$. As d_{t+1} and p_{t+1} are independent (both truly and as perceived),

$$\begin{aligned} \alpha \text{Var}_{t+1}^i(d_{t+2}(k) + p_{t+2}(k)) &= \alpha \text{Var}_t^i(d_{t+1}(k) + p_{t+1}(k)) \\ &= \alpha (V_1(k) + A_{DC}(k)^2 (N^2 \sigma_k^2 + N \tau_k^2)) \end{aligned}$$

The numerator can be simplified. First,

$$\begin{aligned} &E_{t+1}^i[p_{t+2}(k)] - p_{t+1}(k) \\ &= E_{t+1}^i[p_{t+2}(k)] - A_{DC}(k) \sum_i s_{it+1} - B_{DC}(k) \mu_k - E_{t+1}^i[p_{t+2}(k)] + \alpha \text{Var}_{t+1}^i(d_{t+2}(k) | s_{t+1}) \\ &= -A_{DC}(k) \sum_i s_{it+1}(k) - B_{DC}(k) \mu_k + \alpha \text{Var}_{t+1}^i(d_{t+2}(k) | s_{t+1}) \end{aligned}$$

Similarly,

$$E_t^i[p_{t+1}(k)] - p_t(k) = -A_{DC}(k) \sum_i s_t^i(k) - B_{DC}(k) \mu_k + \alpha \text{Var}_t^i(d_{t+1}(k) | s_t)$$

Then, noting that $\text{Var}_{t+1}^i(d_{t+2}(k) | s_{t+1}) = \text{Var}_t^i(d_{t+1}(k) | s_t)$,

$$E_{t+1}^i[p_{t+2}(k)] - p_{t+1}(k) - (E_t^i[p_{t+1}(k)] - p_t(k)) = -A_{DC}(k) \sum_i (s_{it+1}(k) - s_t^i(k))$$

Lastly,

$$\begin{aligned} E_{t+1}^i[d_{t+2}(k)] &= \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} s_{it+1}(k) \\ E_t^i[d_{t+1}(k)] &= \frac{\tau_k^2}{\sigma_k^2 + \tau_k^2} \mu_k + \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} s_t^i(k) \end{aligned}$$

Combining the terms, and noting that the loading of $E_{t+1}^i[d_{t+2}(k)]$ on the trader's *own* signal is $\frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2} = N A_{DC}(k)$, the volume of trade generated by trader i is:

$$v_t(k) = \left| \frac{N A_{DC}(k) (s_{t+1}^i(k) - s_t^i(k)) - A_{DC}(k) \sum_j (s_{t+1}^j(k) - s_t^j(k))}{\alpha (V_1(k) + A_{DC}(k)^2 (N^2 \sigma_k^2 + N \tau_k^2))} \right|$$

Write $s_{t+1}^j(k) - s_t^j(k) = (d_{t+2}(k) - d_{t+1}(k)) + (\varepsilon_{jt+1}(k) - \varepsilon_{jt}(k))$. The common dividend

innovation $d_{t+2}(k) - d_{t+1}(k)$ enters the own-signal term and the sum with the same total weight $NA_{DC}(k)$, so it cancels: common news moves all traders' expectations in parallel and generates no trade. Only idiosyncratic noise remains in the numerator,

$$\begin{aligned} X &\equiv NA_{DC}(k) (\varepsilon_{it+1}(k) - \varepsilon_{it}(k)) - A_{DC}(k) \sum_j (\varepsilon_{jt+1}(k) - \varepsilon_{jt}(k)) \\ &= A_{DC}(k) \left[(N-1) (\varepsilon_{it+1}(k) - \varepsilon_{it}(k)) - \sum_{j \neq i} (\varepsilon_{jt+1}(k) - \varepsilon_{jt}(k)) \right] \end{aligned}$$

which is normal with ex-ante (time $< t$) mean $E[X] = 0$. The noise differences are independent across traders with $\text{Var}(\varepsilon_{jt+1}(k) - \varepsilon_{jt}(k)) = 2\tau_k^2$, hence:

$$\text{Var}(X) = A_{DC}(k)^2 [(N-1)^2 \cdot 2\tau_k^2 + (N-1) \cdot 2\tau_k^2] = 2A_{DC}(k)^2 \tau_k^2 N(N-1).$$

Turning to the ex-ante expectation of the absolute value, and substituting $A_{DC}(k) = \frac{1}{N} \frac{\sigma_k^2}{\sigma_k^2 + \tau_k^2}$ (the factor σ_k^2 cancels between numerator and denominator),

$$E_{<t}[v_t(k)] = \frac{\sqrt{\text{Var}(X)} \sqrt{\frac{2}{\pi}}}{\alpha (V_1(k) + A_{DC}(k)^2 (N^2 \sigma_k^2 + N \tau_k^2))} = \frac{2\tau_k (\sigma_k^2 + \tau_k^2) \sqrt{N(N-1)/\pi}}{\alpha [N (\sigma_k^4 + \sigma_k^2 \tau_k^2 + \tau_k^4) + \sigma_k^2 \tau_k^2]}.$$

For the comparative statics, write $s = \sigma_k^2$, $u = \tau_k^2$, and $Q = s^2 + su + u^2$. Differentiating the logarithm in N ,

$$\frac{\partial}{\partial N} \log E_{<t}[v_t(k)] = \frac{2N-1}{2N(N-1)} - \frac{Q}{NQ + su} = \frac{NQ + (2N-1)su}{2N(N-1)(NQ + su)} > 0,$$

so the expected volume strictly increases with N for every $N > 1$, converging to $\frac{2\tau_k(\sigma_k^2 + \tau_k^2)}{\sqrt{\pi}\alpha Q}$ as $N \rightarrow \infty$. Differentiating in s ,

$$\frac{\partial}{\partial s} \log E_{<t}[v_t(k)] = \frac{1}{s+u} - \frac{N(2s+u) + u}{NQ + su} = -\frac{Ns(s+2u) + u^2}{(s+u)(NQ + su)} < 0,$$

so the expected volume strictly decreases with σ_k , with no restriction on the relative sizes of σ_k and τ_k .

Finally, the time range. The cancellation of the deterministic parts of demand across the two first-order conditions uses the stationarity of the hedging covariance terms in Equation 12, which holds when both trading dates are separated from the final-period boundary, i.e., for $t < T-3$.